

Leveraging Supervised Machine Learning Techniques in Detecting Anomalies
in Food Commodity Prices: Insights for Kenya's Market Stability

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Declaration and Approval

I declare that this work has not been previously submitted and approved for the award of a degree by this or any other university. To the best of my knowledge and belief, the Thesis contains no material previously published or written by another person except where due reference is made in the research itself.

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Abstract

Price volatility in Kenya's food commodity markets has a significant impact on economic stability and food security, posing substantial risks such as financial losses for farmers and traders, as well as increased vulnerability to food shortages for communities. The current manual processes employed to detect anomalies in food commodity pricing are inefficient, slow, and computationally demanding, resulting in delayed and inadequate responses to market disruptions. This presents a critical challenge, as timely detection and intervention are essential for maintaining market stability and ensuring food availability. This research proposes the application of supervised machine learning techniques to systematically identify pricing anomalies. The study aims to evaluate and select the most effective and suitable model for anomaly detection. Leveraging historical commodity pricing data, the study intends to demonstrate how these machine learning approaches can provide a robust alternative to manual analysis. The expected outcomes include the development of a scalable, efficient, and user-friendly cloud-based system capable of real-time anomaly detection and identification. This system will facilitate continuous learning from emerging market data, enabling prompt identification of unusual pricing trends. Consequently, stakeholders will gain immediate and actionable insights, empowering proactive decision-making and timely interventions. The ultimate goal is to substantially enhance Kenya's food market stability and strengthen food security by providing policymakers, traders, and farmers with precise market intelligence.

Keywords: *price volatility, anomaly detection, machine learning, food security, economic stability, supervised learning, Kenya*

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List of Abbreviations

RNN Recurrent Neural Networks

NDVI Normalised Difference Vegetation Index

SPI Standardised Precipitation Index

ALPS Alert for Price Spikes

CPI Consumer Price index

FEWS NET Famine Early Warning Systems Network

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CHAPTER 1: INTRODUCTION

1.1 Background Information

Food price stability is crucial for economic and social well-being, particularly in developing countries like Kenya, where a significant portion of household income is spent on food (Yusuf and Adelabu, 2024). According to Gbashi and Njobeh (2024), anomalies in food prices can have consequences that not only affect food security but also inflation and the overall stability of the food commodity market. Thus, food prices are a critical component in stabilising the market, yet they are subject to significant and often unanticipated fluctuations due to diverse economic, political and environmental changes. These irregularities in price, referred to as anomalies, can potentially destabilise markets, strain household budgets, and exacerbate food insecurity. This, therefore, implies that there is a need for timely identification of price anomalies to inform policy decisions and enable timely interventions to support the market.

A variety of factors contribute to anomalies in food prices in developing countries, including Kenya. A study by Brown and Kshirsagar (2015) highlights international shocks and weather as key factors causing price fluctuations. Weather, as a central component, determines seasonal patterns that affect harvest cycles, which are critical for establishing predictable changes in supply and demand. On the contrary, unexpected disruptions such as drought, floods, geopolitical instability and supply chain breakdowns lead to sudden and unpredictable drops or spikes in food prices (Awange et al., 2007). These price anomalies may have extreme implications that affect the ability of households to access affordable food and threaten the sustainability of agricultural producers.

However, as observed by Ilaha (2022), the traditional methods of monitoring and predicting fluctuations in food prices occasionally fail to capture complex market dynamics and identify subtle anomalies that may indicate emerging trends or potential crises. The author further indicates that relying on traditional statistical methods such as analytical indicators, factor modelling, comparative and grouping techniques, as well as manual monitoring, which are common in foundational economic analysis, is labour-intensive and time-consuming. Equally, as postulated by Pitchaimuthu and Holla (2024), while these methods can provide useful insights, they are often limited in their ability to process large datasets or capture complex patterns, especially when multiple variables interact. Thus, coupled with the growing availability of detailed food price data, there is an urgent need for more advanced and scalable tools that can quickly help detect and interpretation of anomalies.

Therefore, this study proposes machine learning techniques as a solution to addressing these challenges. In contrast with traditional methods, machine learning models are suitably placed because of their capability to process large volumes of data and identify hidden patterns that can reveal price anomalies (Korir et al., 2020). These models can handle non-linear relationships and complex interactions between variables, making them suitable for the intricate dynamics common in food price markets. For instance, deep learning models like recurrent neural networks (RNNs) and long short-term memory (LSTM) networks have demonstrated particular advantages in capturing temporal dependencies in time series data (Santoro et al., 2024). Similarly, studies have demonstrated that machine learning algorithms like eXtreme gradient boosting (XGBoost) have often performed better than the RNN-LSTM model for predicting accuracy compared to other shallower algorithms using stationary data.

1.2 Problem Statement

Kenya's food commodity markets are characterised by frequent and uneven price fluctuations, influenced by a range of factors including seasonal production cycles, climate variability, transport constraints, policy shifts and global commodity market dynamics (Nyoro et al., 2004; FAO, IFAD, UNICEF, WFP and WHO, 2022). While some of these price shifts reflect genuine market shocks, such as droughts, flooding or disruptions in supply chains, others are a result of anomalies in data collection, reporting errors or temporary localised events (Gbadegesin et al., 2024).

However, the existing monitoring systems typically depend on manual data inspection or simple rule-based thresholds applied to aggregated datasets. These approaches often fail to capture subtle yet quite important deviations, and they are also slow in generating alerts and likely to often produce false positives and false negatives (Gustafson, 2013). This undermines timely decision-making with negative impacts for policymakers, traders and humanitarian agencies to implement targeted interventions for stabilising prices and maintaining market resilience. The study proposes supervised machine learning techniques to support the development of the capacity to learn complex, non-linear patterns in historical price data, accounting for seasonality, spatial heterogeneity and inter-commodity relationships more effectively than conventional statistical methods (Jordan and Mitchell, 2015; Aggarwal, 2017).

1.3 Research Objectives

1.3.1 General Objectives

This study aims to develop a supervised machine learning framework for detecting anomalies in food commodity prices to strengthen Kenya's market monitoring and stability.

1.3.2 Specific Objectives

The following specific objectives will be adopted in implementing this study;

- i) To develop a data engineering process that captures seasonality, spatial variation and inter-commodity relationships in Kenyan food commodity price data.
- ii) To train supervised machine learning models for anomaly detection.
- iii) To evaluate the scalability of the developed framework for application to similar market contexts.

1.4 Research Questions

The following research questions will be adopted to achieve the objectives of this study;

- i) How can data engineering be designed to integrate seasonality, spatial variation and inter-commodity relationships in the Kenyan food commodity price data?
- ii) How can supervised machine learning models be trained to detect anomalies in food commodity prices?
- iii) What is the scalability of the developed ML framework for application in similar market contexts?

1.5 Significance and Justification of the Study

This study addresses an existing problem in Kenya's market monitoring capacity by developing a supervised machine learning framework capable of detecting anomalies in food commodity prices with accuracy and speed than current manual or threshold-based approaches ([Jordan and Mitchell, 2015](#); [Aggarwal, 2017](#)). These improvements can enable policymakers, regulatory bodies and humanitarian agencies to respond effectively to real market disruptions, reducing the risk of prolonged instability in the food market. The study also advances academic understanding by demonstrating how tailored feature engineering, like capturing seasonality, spatial variation and inter-commodity relationships, can enhance model performance in agricultural price analysis ([Newton et al., 2021](#)). Most importantly, however, the framework design is meant to be architecturally scalable, which is critical to ensure adaptability with different datasets and market contexts with similar characteristics beyond Kenya ([Mehendale, 2024](#)).

1.6 Definition of Key Terms

The following key terms: food commodity and food prices are critical to the scope and methodology of this study. Therefore, providing a clear definition will help establish a shared understanding of both the technical and economic dimensions of this study, enabling proper alignment of the findings.

1.6.1 Food Commodity

For this study, a food commodity refers to raw agricultural products that are traded, processed or consumed in their primary form ([Fontefrancesco, 2019](#)). The examples here include staple cereals, legumes, root crops and unprocessed animal products, which form the foundation of national and global food systems.

1.6.2 Food Prices

Food prices, as defined by [de Janvry \(1982\)](#), are the monetary values assigned to agricultural and food commodities, which influence real incomes, wages and consumption patterns within a population. They are regarded as essential wage goods that shape employment, rural welfare and political stability. [McCalla \(2009\)](#) distinguishes between nominal prices, measured in current monetary terms and real prices, adjusted for inflation to reflect purchasing power.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing studies on the application of supervised machine learning techniques for detecting anomalies in food commodity price data. It examines how different algorithms have been used to identify irregular patterns, capture market dynamics and improve the accuracy of price monitoring. The review also highlights gaps in current approaches that the present study seeks to address,

2.2 Empirical Literature Review

2.2.1 Seasonality, Spatial and Inter-Commodity Feature Engineering in Food Prices

Food commodity markets are characterised by strong and predictable seasonal cycles tied to production, harvest and climate shocks. Empirical studies highlight the importance of accounting for these intra-year fluctuations using decomposition techniques like Seasonal-Trend decomposition using Loes (STL), Fourier terms and calendar dummies ([Gilbert et al., 2017](#); [Zhang et al., 2025](#)). However, [Gilbert et al. \(2017\)](#) study on food price seasonality in Africa demonstrated that dummy-based seasonality often underestimates the amplitude of fluctuations, whereas the trigonometric and sawtooth techniques are more effective in capturing sharp trough-to-peak gaps that are commonly observed in African staple markets. In the context of Kenya, maize prices are a clear example which usually fluctuates by nearly one-third across harvest cycles.

Beyond cyclical dynamics, recent studies emphasise that agro-ecological signals substantially enhance the predictive capacity of food price models. Weather-based features such as rainfall anomalies, the Standardised Precipitation Index (SPI), and the Normalised Difference Vegetation Index (NDVI) have been widely integrated to capture production shocks and explain market volatility. A study by [Shuaibu et al. \(2016\)](#) found that NDVI anomalies strongly aligned with drought episodes and corresponded with significant maize price surges in Kenyan markets, underscoring the role of vegetation health as a leading indicator of price instability. Supportive evidence from Niger confirmed that reduced rainfall intensifies intra-seasonal price variability, often exerting a delayed effect on staple food availability and affordability.

Further, spatial heterogeneity, influenced by transport costs, cross-border trade, and infrastructural constraints, plays a critical role in shaping Kenya's food commodity price differentials. For example, [Kihiu and Gachanja \(2022\)](#) found that improved rural road infrastructure is associated with lower maize prices in Kenyan retail markets and that spatial price dependence exists across neighbouring counties affected by transport networks. Equally, spatial economet-

ric tools like Moran's I and network-based methods have been used to test market integration and highlight the role of trade corridors in price transmission. For instance, [Abdulai \(2000\)](#), [Waiswa \(2023\)](#) and [Waiswa and Yavuz \(2023\)](#) study in Ghana and East Africa, respectively, employed cointegration tests and nonlinear ARDL models, which showed strong cross-border price transmission between Nairobi and Uganda markets, indicating that a 1% increase in maize prices in Nairobi leads to nearly a 0.9% increase in Ugandan prices. Such evidence illustrates that integrating spatial data, based on geographical distance, road connectivity, and neighbouring market price lags into the supervised machine learning models can bring meaningful improvement in detecting price anomalies.

Moreover, inter-commodity relations are critical in understanding the dynamics of commodity markets, as they often co-move under shared shocks. [Kabundi and Zahid \(2023\)](#) study highlighted the existence of global and group cycles in commodity markets that are largely driven by global supply shocks since 2000, which have increased the demand for commodities like grains and precious metals. Similarly, [Esposti \(2017\)](#)'s research supports the presence of a common latent factor that affects commodity price movements, with significant long-term equilibrium price increases observed in recent decades. However, regional studies revealed something different. In the Indian context, [Kumar \(2006\)](#) analysis revealed unique long-term relationships among certain cereals, pulses and oilseeds. Here, government policy instruments such as minimum support prices are provided as stabilisers against disturbances that might otherwise transmit freely into the commodity market. This raises questions as to whether the observed co-movements are truly global or are mediated by local policy frameworks. Other researchers have shifted the lens towards energy-food linkages. [Mensi et al. \(2014\)](#) highlights the dynamic spillovers between energy and cereal prices, where volatility in oil markets amplifies fluctuations in grain markets. Similarly, [Kozian et al. \(2025\)](#) tested methodological boundaries by comparing VAR and VARX models to machine learning approaches like Random Forests, ultimately finding that the econometric models capture cross-commodity dynamics more effectively.

2.2.2 Comparing Supervised ML Models against Threshold-based Monitoring

Studies on food price monitoring have tested supervised machine learning approaches against traditional threshold-based methods. [Minot \(2014\)](#) and [Food and Agriculture Organization of the United Nations \[FAO\] \(2025\)](#) studies demonstrate that rule-based z-scores and interquartile thresholds are often inadequate in volatile contexts, as they usually fail to capture regime shifts

resulting from trade shocks and drought. This shortcoming has motivated the adoption of supervised machine learning methods that can better adapt to structural changes in price dynamics.

However, since labelled anomalies are rare in food market data, many studies have employed the use of synthetic anomaly generation. For instance, [Ramírez et al. \(2020\)](#) proposed a latent-space approach that distils inlier representations and synthesises pseudo-outliers from the outlier distribution, a method that enables robust classifier training even without notable examples of anomalies in data. Broadly, [Jiang et al. \(2023\)](#) reviewed weakly supervised anomaly detection techniques suitable for time-series and tabular data and presented techniques like incomplete, inexact and inaccurate supervision that align well with the challenge of food price anomaly detection labelling. Equally, researchers have proposed a combination of techniques like focal loss, which emphasises difficult samples over easy ones ([Lin et al., 2018](#)) and cost-sensitive learning, where loss function is weighted to disadvantage mistakes on minority classes ([Baloch et al., 2019](#); [Ahmad and Rahimi, 2025](#)). As such, methods like Focused Anchor Loss exhibit the effectiveness of combining these approaches ([Baloch et al., 2019](#)), generally to cope with class imbalance that characterises the classification of anomalies.

A diverse set of literature demonstrates that supervised machine learning approaches like support vector machines, decision trees and gradient boosting outperform conventional threshold-based monitoring methods in anomaly detection. [Minot \(2014\)](#) criticises threshold methods for their inability to capture complex, nonlinear market dynamics and structural breaks. This is unlike the supervised machine learning models that leverage historical anomaly patterns and contextual features to identify both sudden shocks like drought-induced maize price spikes and slower, structural distortions tied to transport challenges or cross-border trade disruptions ([Minot, 2014](#)).

2.2.3 Scalability and Transferability of ML Model

A growing body of literature emphasises the importance of designing scalable models that adapt to new network settings and data environments.

[Andrée and Pape \(2023\)](#) study examined whether machine-learning models can fill gaps in market-level food price series across multiple fragile settings. Using partial survey data from over 1,200 markets in 25 countries, the approach trained commodity-specific models that impute missing prices from nearby markets and related items. In validation against held-out observations, the method captured about 85% of observed price variation even when 60% to 80% of survey data were missing, indicating a cross-country scalability for real-time monitoring.

However, the study noted that in fragile contexts where both price data and covaries are thin, it may not be possible to determine transferability due to instability in market linkages. The study, therefore, emphasised that while scalable, models still require a minimum density of observation to maintain accuracy.

Equally, [Baquedano \(2015\)](#) developed the FAO's Indicator of Food Price Anomalies (IFPA), which applied z-score thresholds to growth rates for anomaly detection. The method provided a standardised anomaly label that supports supervised machine learning in cross-country settings. The authors concluded that IFPA enhances transferability by harmonising anomaly definitions. However, [Tadesse et al. \(2014\)](#) criticism such statistical anomaly frameworks for overlooking structural weaknesses of volatility, such as cross-border trade policies or speculative dynamics, which weaken predictive robustness when applied in new contexts.

A study by the [World Food Programme \(2014\)](#) developed the ALPS (Alert for Price Spikes) indicator, which measures deviations from seasonal price expectations. ALPS has been praised for its operational utility, as it has been integrated into multiple country dashboards and reused to train supervised anomaly detectors. The findings concluded that ALPS is the best suited as a real-time trigger tool. However, critics aver that ALPS design is less suited to capturing gradual or sustained price increases that evolve over longer periods. A recent study by [Busker et al. \(2024\)](#) notes that while ALPS effectively identifies short-lived shocks, it tends to underperform when faced with slow-moving or cumulative price rises, thereby limiting its relevance for long-term policy planning and structural food market assessments. These findings suggest that ALPS is best applied as a complementary tool alongside broader analytical frameworks that track persistent market trends.

Further, [Herteux et al. \(2024\)](#) applied a reservoir-computing (RC) model to forecast food security trends in Mali, Nigeria, Syria and Yemen. The RC approach achieved a 5 percentage-point RMSE for 60-day forecasts and 0.43 accuracy on deterioration classification. Besides, the findings showed that the RC models performed better in forecasting food security trends than conventional methods like ARIMA and LSTM, particularly when extended beyond 5 forecast steps, maintaining lower error growth and offering more reliable predictions in deteriorating conditions. However, the ability of RC to identify rare but severe events remains limited, since these cases are underrepresented in the training data. Similar concerns are reflected in the findings of [Busker et al. \(2024\)](#), who noted that machine learning systems trained for broad cross-country applications often lose sensitivity to extreme shocks. These studies thus demonstrate

that whereas RC enhances overall forecasting stability, it is still limited in detecting high-impact crises.

In addition, [Foini et al. \(2023\)](#) employed gradient-boosted models across six countries, achieving 99% one-day accuracy in Syria but dropping to 47% at 30 days in Yemen. The authors concluded that transferability is strongest for short horizons, but weak for longer-term warnings without retraining. Independent evaluations of early-warning systems support this limitation, noting that while such models achieve high overall accuracy, they often underperform in predicting the most severe crises, leading to missed or delayed alerts for high-impact events [Maxwell et al. \(2019\)](#). This suggests that while gradient-boosted approaches enhance short-term forecasting, their reliability in crisis-prone environments remains constrained.

2.2.4 Gaps in Related Works

A review of related works reveals three major emerging gaps. First, the existing works have not fully integrated climatic, spatial and inter-commodity features into supervised ML pipelines ([Jordan and Mitchell, 2015](#); [Minot, 2014](#); [Aggarwal, 2017](#)). This issue constrains the ability of the developed models to capture the complex drivers of food commodity price volatility fully. Second, anomaly labelling remains a challenge, often relying on synthetic data integration that risks deviating from real-world dynamics, thereby weakening the reliability of trained classifiers ([Mehendale, 2024](#); [Wang and Zhao, 2023](#)). Third, the scalability and transferability of models remain uncertain, especially in fragile markets where rare but extreme shocks are dominant ([Aggarwal, 2017](#); [Newton et al., 2021](#)). As a result, this study addresses the underlying gaps by developing supervised ML models that incorporate richer feature sets, apply advanced methods for handling class imbalance and test scalability in Kenyan food commodity markets with potential for regional application.

CHAPTER 3: METHODOLOGY

This study applied the Cross-Industry Standard Process for Data Mining (CRISP-DM) as the methodological framework. CRISP-DM is widely recognised in applied data science and offers a structured yet iterative process that links problem definition to deployment (Wirth and Hipp, 2000; Durango et al., 2023). Its modular design is particularly relevant for anomaly detection in food commodity markets, where heterogeneous datasets and structural shocks complicate analysis (Food and Agriculture Organization of the United Nations [FAO], 2025). The stages followed in this study were data understanding, collection, cleaning, model development, performance evaluation, and deployment, as shown in Figure 3.1.

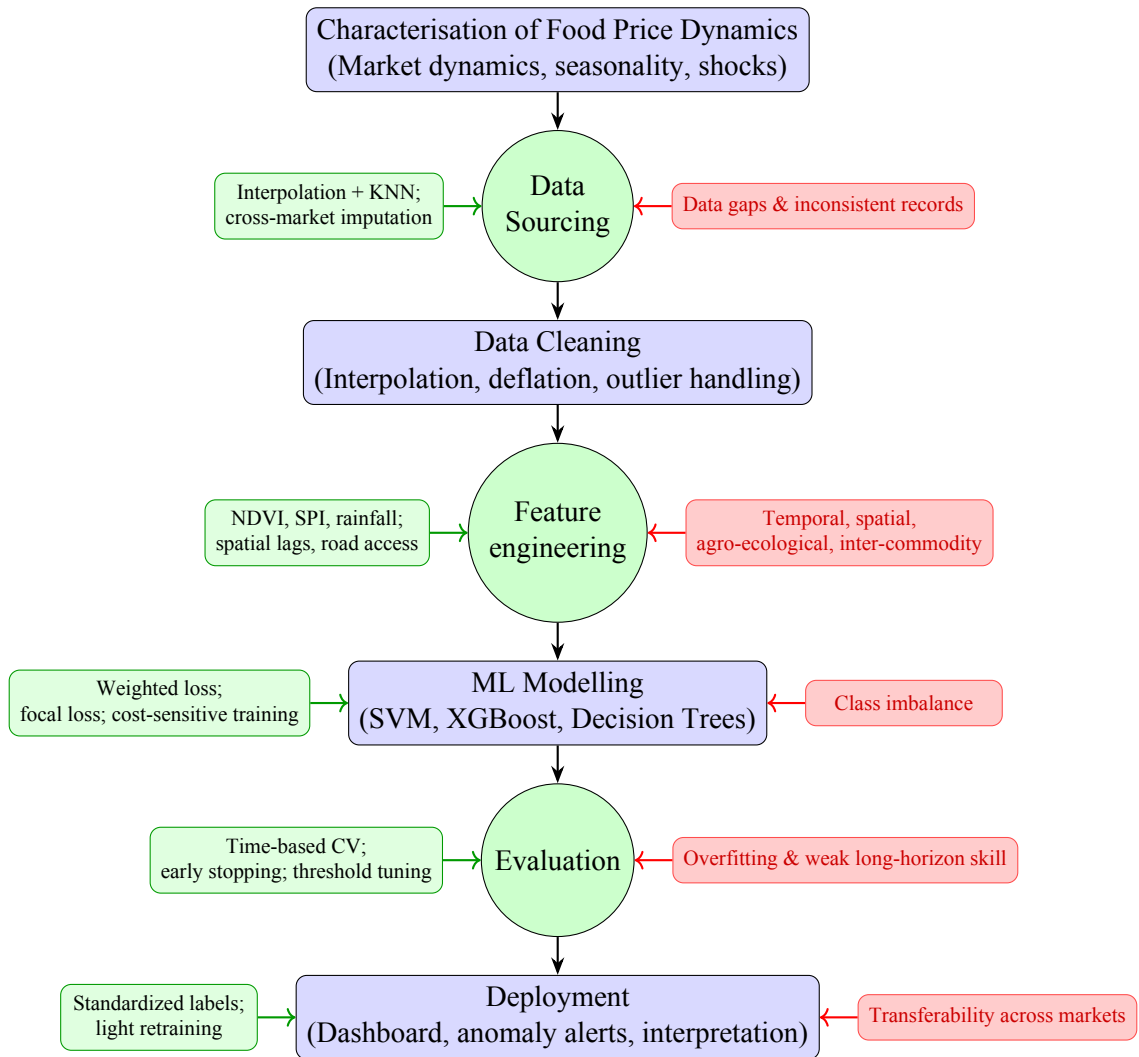


Figure 3.1: Methodology workflow with stages (rectangles), checkpoints (circles), challenges (red), and solutions (green) for supervised ML anomaly detection in food commodity prices.

The overall aim of this methodological framework was to develop a scalable anomaly detection system capable of providing early warnings of abnormal food price movements and

supporting stakeholders in maintaining market stability.

3.1 Characterisation of Food Price Dynamics

In this study, the characterisation stage involved establishing the structural properties of food commodity price series in Kenya by analysing their patterns, spatial dependencies and exposure to irregular shocks. Historical data on staples such as maize, beans and rice were examined for recurrent harvest-linked cycles using decomposing techniques to isolate trend and seasonal components, as recommended in previous studies on African food price volatility ([Gilbert et al., 2017](#); [Minot, 2014](#)). Further, spatial linkages were assessed by incorporating market connectivity indicators, such as distance to Nairobi and proximity to cross-border trade routes, which have been shown to significantly influence price transmission across counties and neighbouring countries ([Kihui and Gachanja, 2022](#); [Waiswa, 2023](#)). In addition, climate variability was captured using rainfall anomalies and NDVI indices, reflecting the evidence that agro-ecological signals strongly predict drought-induced price surges ([Shuaibu et al., 2016](#)). This characterisation provided the basis for defining anomalies not as random outliers, but as deviations from expected price trajectories once seasonal, spatial and climatic dynamics are accounted for. This was intended to ensure that the subsequent modelling is both data-driven and context-sensitive.

3.1.1 Data Sourcing

The dataset for this study was primarily collected from the World Food Programme (WFP) Price Database via the [OCHA Services](#) platform for the period of two hundred and thirty-eight (238) months, starting 15 January 2006 to 15 November 2025. This source provided monthly retail and wholesale prices of major staples, including maize, beans and rice, across different Kenyan markets. The primary data was complemented with official consumer price index and market-level retail statistics data sourced from the Kenya National Bureau of Statistics (KNBS) to ensure consistency with national economic indicators. The [FAO Food Price Monitoring and Analysis \(FPMA\) tool](#) provided a sample of international reference prices and comparative regional data to facilitate checks and contextualization of domestic price movement. At the same time, [FEWS NET market bulletins](#) was used to extract early warning price data and qualitative assessments of shocks such as droughts, trade restrictions and supply chain disruptions.

Further, the study incorporated auxiliary datasets to capture non-price volatility drivers. These included rainfall anomalies and the SPI for climatic variability, the NDVI for crop and pasture conditions and infrastructure indicators such as road density, distance to Nairobi and proximity to trade corridors for spatial dependencies. The integration of WFP market data with

official statistics, international reference series and agro-ecological variables provided broad temporal coverage and contextual depth, reflecting recommendations from previous work on food price modelling (Minot, 2014; Shuaibu et al., 2016). Table 1 summarises the data sources and type of information that the study obtained and utilised in the implementation of this project.

Table 1: Data sources and type of information collected for the study

Source	Type of Data Collected	Purpose/Use in Study
World Food Programme (WFP) Price Database via OCHA Services	Monthly retail and wholesale food prices for major staples (maize, beans, rice) across Kenyan markets	Main dataset for anomaly detection; provided consistent, harmonised price series across regions
Kenya National Bureau of Statistics (KNBS)	Consumer price indices (CPI) and official retail market prices	Adjusted for inflation and aligned local price series with national economic indicators
FAO Food Price Monitoring and Analysis (FPMA) Tool	International and regional reference food prices	Enabled comparison of Kenyan prices with global and regional benchmarks; supported cross-checks
FEWS NET Market Bulletins	Early warning market prices and qualitative assessments of shocks (e.g., droughts, trade bans, supply chain disruptions)	Provided context on external shocks and qualitative explanations for abnormal price behaviour
Auxiliary Datasets (Climate and Infrastructure)	Rainfall anomalies, Standardised Precipitation Index (SPI), Normalised Difference Vegetation Index (NDVI), road density, distance to trade hubs	Captures agro-ecological and spatial factors affecting price dynamics and transmission

3.1.2 Data Cleaning and Preprocessing

The collected datasets were preprocessed to ensure they were suitable for a supervised ML model for anomaly detection. The missing observations in short intervals were treated using linear interpolation. At the same time, longer gaps were addressed using K-nearest neighbour (KNN) imputation, which leverages correlations between neighbouring markets, as applied in recent market integration studies by Waiswa (2023) and Waiswa and Yavuz (2023). This technique estimates the missing observations by identifying similar data points within the dataset and averaging their values. Compared to simple imputation, KNN imputation preserves the local structure of the data and reduces bias. Outlier screening was performed to remove clear reporting errors, but genuine spikes were retained since they represent anomalies of interest. All prices were converted to real terms using the CPI and a log transformation to stabilise vari-

ance. These steps ensured that the resulting dataset was clean and harmonised, statistically stable and suitable for the supervised machine learning modelling while preserving the underlying economic signals necessary for developing a robust anomaly detection system. Table 2 summarises the processing procedures applied to the raw datasets.

Table 2: Data Cleaning and Preprocessing Procedures

Step	Method	Purpose	Outcome
Missing values (short gaps)	Linear interpolation	Estimate missing observations occurring within short time intervals in the time series	Maintains continuity in the temporal price data
Missing values (long gaps)	K-Nearest Neighbour (KNN) imputation	Estimate missing values using similar observations from neighbouring markets based on spatial correlations	Preserves local data structure and reduces imputation bias
Outlier screening	Visual and statistical inspection	Identify clear reporting errors and unrealistic price entries while preserving genuine market spikes	Erroneous records removed while meaningful anomalies retained
Price adjustment	CPI deflation	Convert nominal food prices into real prices to remove inflationary effects and allow temporal comparison	Harmonised price series across time periods
Data transformation	Log transformation	Stabilise variance and reduce skewness in price distributions for machine learning models	Improved model stability and learning efficiency

3.1.3 Exploratory Data Analysis

The exploratory data analysis (EDA) was undertaken to reveal the structural properties of the Kenyan food price series and assess their suitability for supervised anomaly detection. The main objective here was to identify major patterns, trends and irregularities in the data that influence the modelling processes. This exploration involved visual inspection using time-series plots, decomposition charts and heatmaps to detect seasonal cycles, long-term trends, and sudden disruptions, in line with established approaches to commodity market analysis (Gilbert et al., 2017; Minot, 2014).

Overview of the Dataset. An overview of the dataset was conducted to understand the structure of the dataset, which was necessary to ensure that it is suitable for modelling and for identifying the main characteristics that may influence the analysis. As illustrated in Table 3, the dataset contains 120,510 monthly price observations covering 41 commodities across 225 markets in 27 counties in Kenya, over a period of 238 months.

An examination of these dimensions provided insights into the size of the dataset, geographic coverage and temporal span, which were necessary for capturing spatial market variation and long-term price dynamics.

Table 3: Summary of the WFP Food Commodity Price Dataset for Kenya

Attribute	Description
Number of observations	120,510 monthly price records
Price frequency	Monthly observations
Commodities analysed	Bananas, Beans, Beans (dolichos), Beans (dry), Beans (kidney), Beans (mung), Beans (rosecoco), Beans (yellow), and other staple commodities (41 in total)
Number of commodities	41
Markets covered	225 markets across Kenya
Counties covered	27 counties
Time period	2006–2025
Variables	county, commodity, date, price_nominal, year, month, CPI_index, price_real, log_price, growth_rate, rolling_volatility

Descriptive Statistics of Food Commodity Prices. Table 4 presents the descriptive statistics of the main variables that were adopted in the study, including nominal prices, inflation-adjusted prices and their logarithmic transformations. The results indicated substantial variation in food commodity prices across Kenya markets and over time. The nominal prices have an estimated mean of 1407.49 with a wide range of 5 to 18,000, denoting significant differences in price levels across commodities and periods.

After adjustment for inflation using the CPI index, the real price series also exhibited high variability, reflecting changes in purchasing power and market dynamics. The logarithmic transformation of prices reduces the influence of extreme values and stabilises variance, making the data more suitable for machine learning and modelling. Thus, these statistics provide an overview of the scale, dispersion and distribution of the price variables used in the anomaly detection framework.

Table 4: Summary Statistics of Key Variables

Variable	Count	Mean	Std	Min	25%	Median	75%	Max
price_real	120,510	59.81	53.33	3.29	35.26	48.97	67.94	770.56
NDVI_anomaly	49,560	0.00	0.06	-0.20	-0.04	-0.01	0.04	0.26
rain_mm	49,560	72.65	84.47	0.00	12.75	42.80	104.68	634.16
SPI3	48,734	0.00	1.00	-2.00	-0.81	-0.20	0.48	4.00
SPI6	47,495	0.00	1.00	-2.60	-0.71	-0.21	0.52	3.41
growth_rate	119,995	1.06	11.83	-81.10	-2.69	0.34	3.65	705.14
rolling_vol	119,480	7.32	9.46	0.01	3.34	5.40	8.43	303.80

Temporal Patterns in Food Commodity Prices. Figure 3.2 illustrates the temporal behaviour of selected staple commodities, highlighting recurring seasonal patterns and abrupt price fluctuations. As shown in the left panel, the average monthly seasonal cycles for prices of commodities like beans, rice and sorghum tend to peak during the early months of the year and gradually decline toward the harvest periods, reflecting natural supply-demand dynamics in agricultural markets. Maize shows relatively lower price levels but still follows a similar seasonal pattern. The right panel presents the real price series with sudden spikes identified where month-on-month price changes exceed 30%. There are several spikes exhibited during earlier years of the dataset, suggesting periods of market stress or supply disruptions. These observations confirmed that the dataset contains both predictable seasonal behaviour and irregular price movements, which motivated the use of machine learning models to detect anomalies in food commodity prices.

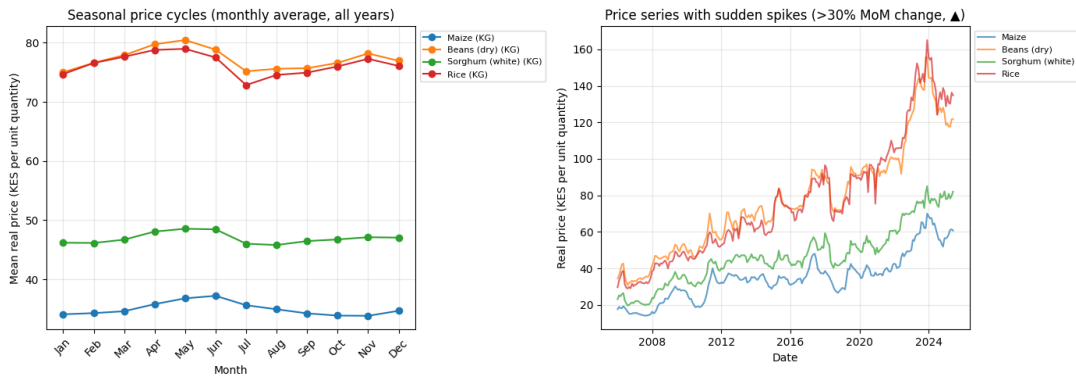


Figure 3.2: Seasonal price patterns and sudden price spikes in selected food commodities.

Seasonal Decomposition of Food Commodity Prices Figure 3.3 presents the Seasonal-Trend decomposition using Loess (STL) decomposition of the national maize price series, illustrating how the observed price dynamics can be separated into trend, seasonal and residual com-

ponents. The trend component captures the long-term evolution of maize prices, while the seasonal component reflects periodic price fluctuations linked to harvest cycles and seasonal supply changes. The residual component highlights irregular deviations from expected price patterns, corresponding to market disruptions or abnormal price movements. The decomposition of the series helped isolate the underlying structure of the food price data and provide a basis for identifying anomalies.

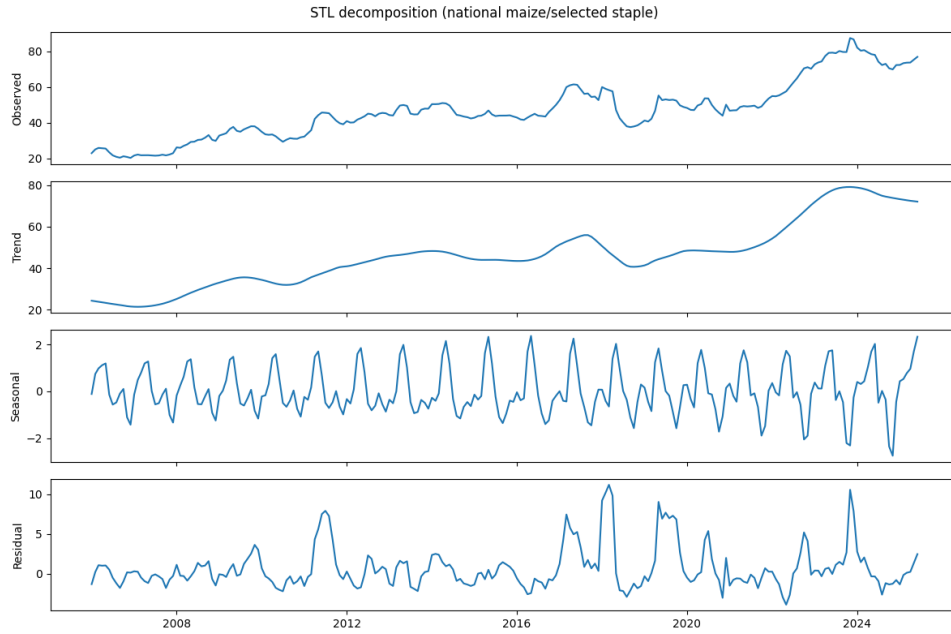


Figure 3.3: Seasonal-Trend decomposition using Loess (STL) applied to the national maize price series.

Price Growth Rate and Volatility Analysis. Growth rates were computed using the formula in Equation 1 to quantify short-term price movements:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100, \quad (1)$$

where r_t denotes the period-to-period price change. Volatility was then captured using rolling standard deviations of returns as shown in Equation 2:

$$\sigma_t = \sqrt{\frac{1}{k} \sum_{i=0}^{k-1} (r_{t-i} - \bar{r})^2}, \quad (2)$$

with k representing the chosen rolling window length, allowing detection of instability during shocks such as droughts or trade restrictions.

Growth rate variables provided additional insight into short-term price volatility by measuring the month-to-month percentage change in commodity prices. Figure 3.4 shows the growth rate series for selected commodities in the Nairobi market. The figure highlights the percentage change in prices over time for beans, kale, maize, aromatic rice, and spinach. Whereas most observations showed stability in prices, several extreme spikes were visible, particularly around the mid 2020s. These spikes indicate abrupt price adjustments.

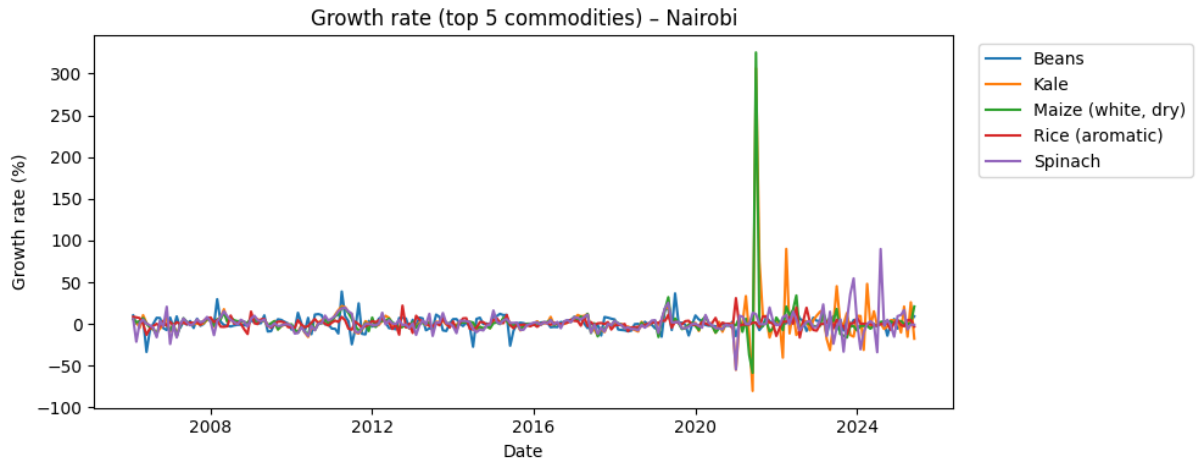


Figure 3.4: Month-on-month price growth rates for selected commodities in Nairobi.

These observations were further illustrated in Figure 3.5, which shows the month-on-month growth rate of kale prices (upper panel) in the Nairobi market alongside the rolling volatility computed over a six-month window ($k = 6$, lower panel). These observations reflected the likelihood of market shocks, supply disruptions or demand fluctuations. Measuring growth rate and rolling volatility helped capture abnormal price behaviour and supported their inclusion as features in the machine learning framework for anomaly detection.

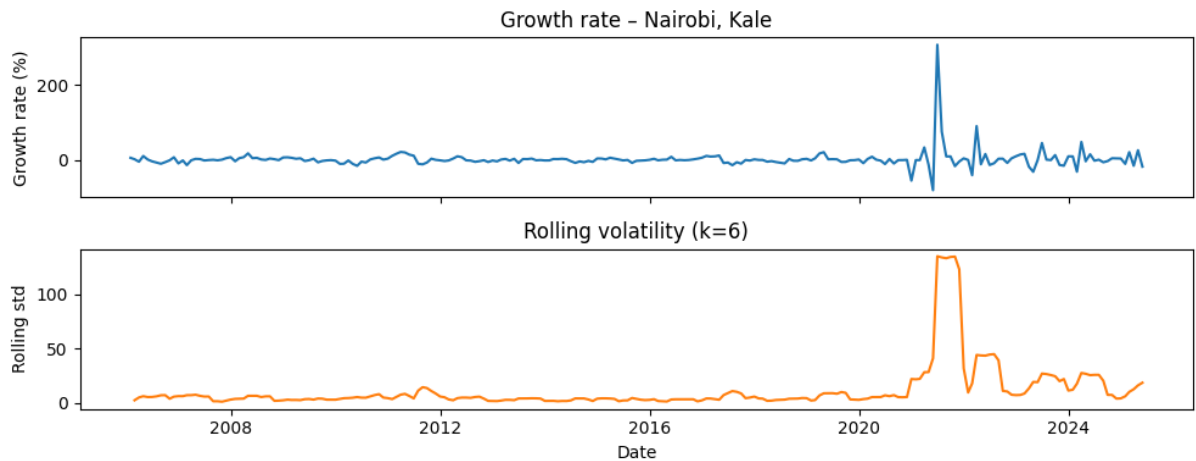


Figure 3.5: Growth rate and rolling volatility for kale prices in the Nairobi market.

Spatial Variations in Market Prices Figure 3.6 is an exhibition of the spatial distribution of mean real community prices across Kenyan counties. The map illustrates the average inflation-adjusted price levels observed across markets, with darker shades indicating higher mean prices. Counties with missing or insufficient observations are indicated with hatched shading.

The map reveals regional variation in average price levels, with some counties showing significantly higher prices than others. These differences reflect variations in market access, transportation costs, climatic conditions and regional supply dynamics. The spatial heterogeneity observed in the data demonstrates the need to account for regional market characteristics when analysing food price dynamics and detecting anomalies.

Moreover, Figure 3.7 shows the regional distribution of mean real commodity prices across Kenya. The results demonstrate noticeable differences in average price levels between regions. Nyanza and the Coastal exhibit relatively higher mean prices, while regions such as Nairobi and the North Eastern showed comparatively lower average price levels. These differences likely reflect variations in agricultural productivity, infrastructure development, transportation costs, and proximity to major consumption markets. Equally, the presence of distinct regional clusters indicates that food price dynamics vary geographically, reinforcing the importance of incorporating spatial context when analysing price behaviour and detecting anomalies.

Spatial Distribution of Mean Commodity Prices Across Kenyan Counties

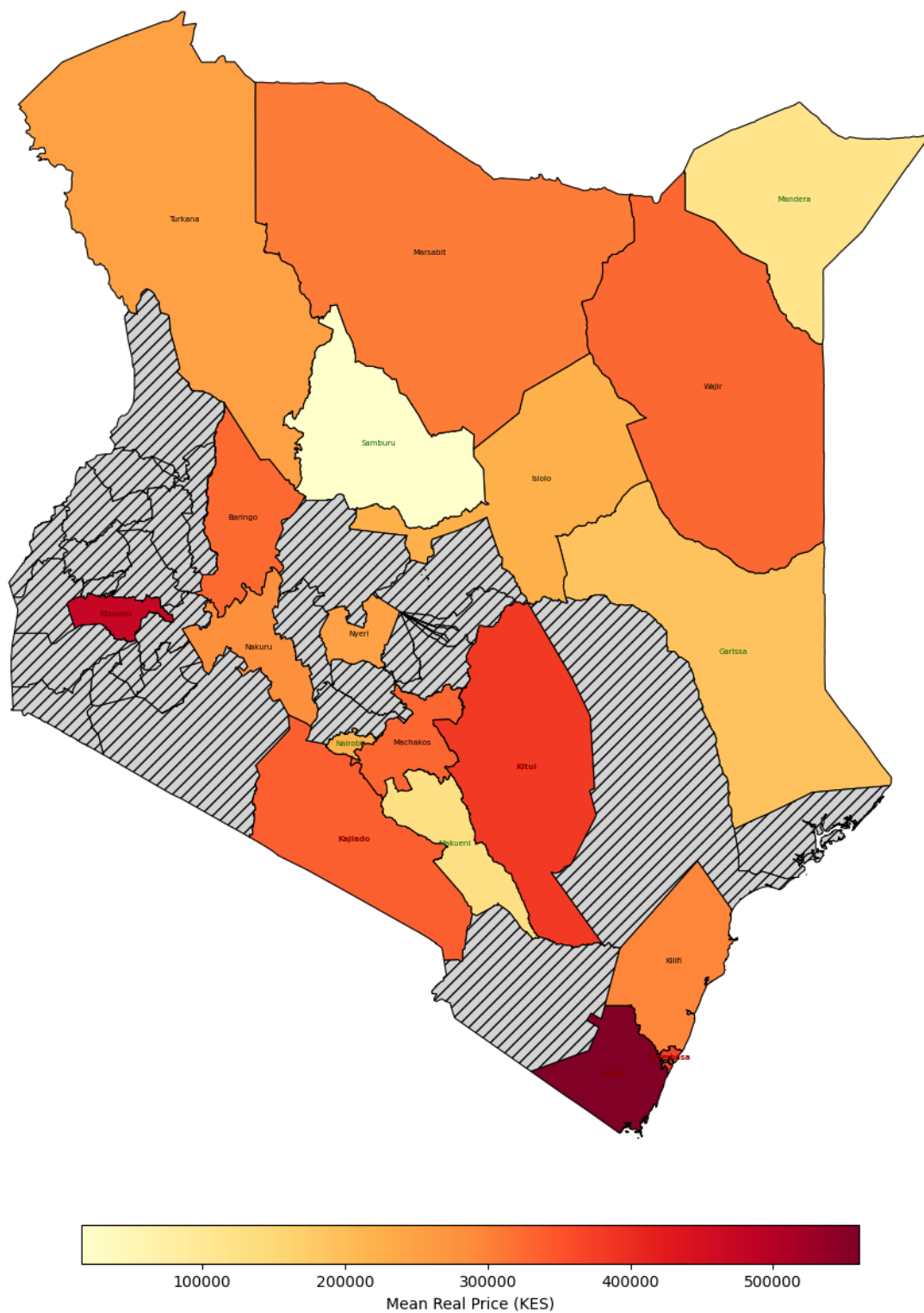


Figure 3.6: Spatial distribution of mean real commodity prices across selected Kenyan counties.

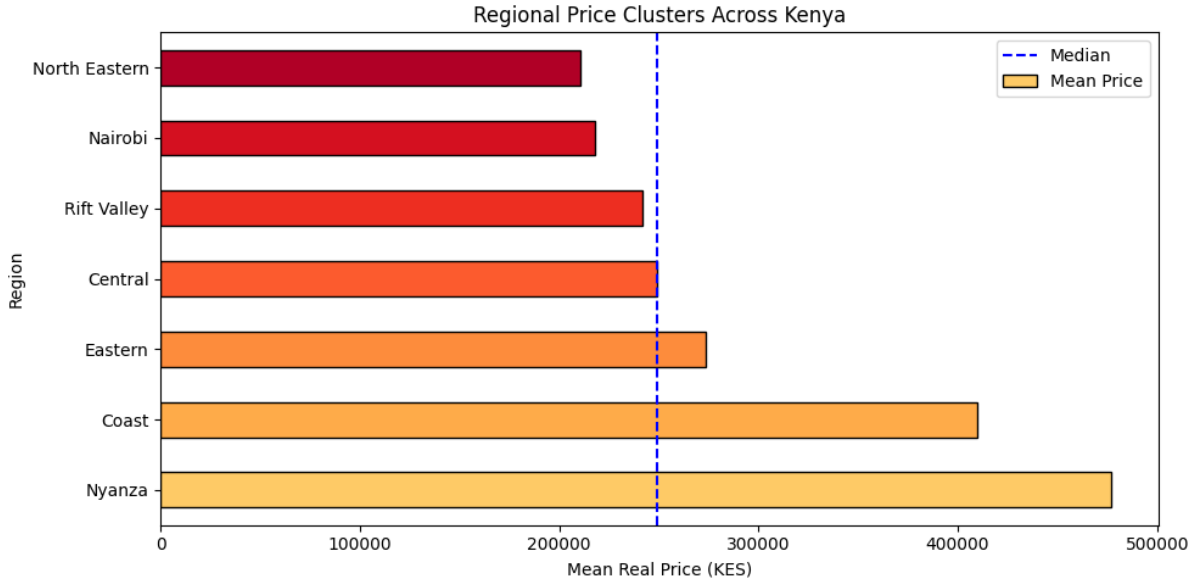


Figure 3.7: Regional clustering of mean real commodity prices across Kenya.

Climate-Price Linkages and Correlation Analysis Subsequently, spatial and agro-ecological linkages were examined through correlation analysis, expressed as shown in Equation 3:

$$\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad (3)$$

to measure the co-movement between market prices and explanatory variables such as rainfall anomalies of NDVI (Shuaibu et al., 2016). These diagnostic steps enabled the identification of candidate anomalies, highlighting the influence of seasonal and climatic drivers.

Figure 3.8 examined the relationship between agro-ecological conditions and food price anomalies across Kenyan counties. Panel (a) presents the correlation matrix between key agro-ecological variables (NDVI anomaly, rainfall, and standardised precipitation indices) and price-related variables. Panels (b), (c), and (d) illustrate the relationships between climate indicators and the rate of detected price anomalies across counties.

The correlation matrix highlights interactions between vegetation conditions, rainfall patterns and price-related variables. The scatter plots further show how variations in NDVI anomalies, rainfall levels and standardised precipitation indices correspond to differences in the rate of price anomalies. Particularly, rainfall exhibits a relatively strong association with anomaly rates, suggesting that climatic variabilities can influence food price instability. These results provided empirical evidence that environmental factors play an important role in shaping food market dynamics, supporting the inclusion of climate indicators in the anomaly detection modelling framework.

Spatial and Agro-Ecological Linkages: Correlation Analysis

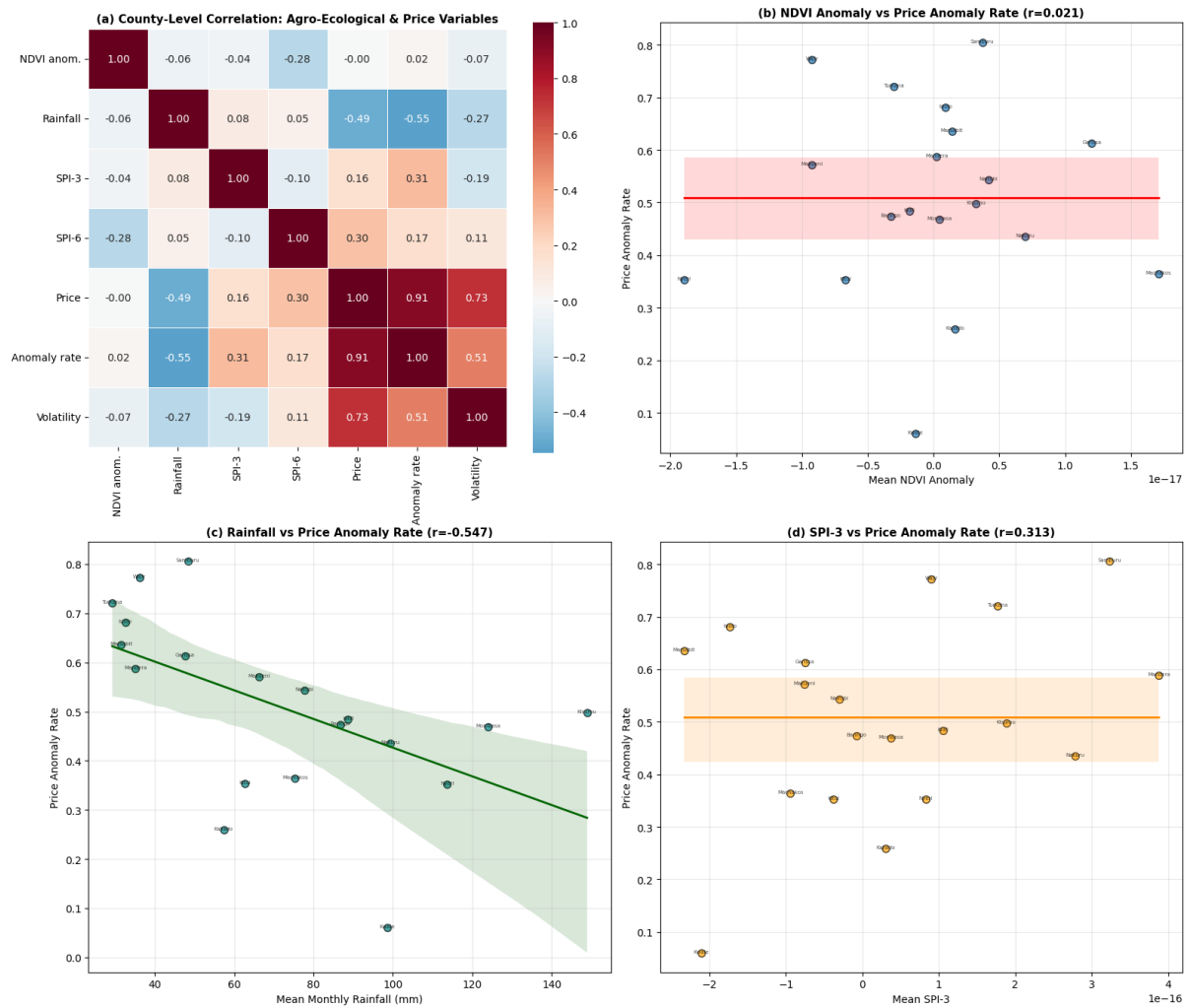


Figure 3.8: Spatial and agro-ecological linkages between climate indicators and food price anomalies.

Table 5 is a summary of the county-level correlations between agro-econological indicators and food price anomaly rates. It is meant to help interpret the charts further.

Table 5: County-Level Correlations Between Agro-Ecological Indicators and Price Anomalies

Variable Pair	Pearson r	Interpretation
NDVI anomaly ↔ Anomaly rate	-0.225	Negative relationship indicating that vegetation loss associated with drought conditions tends to coincide with higher food price anomaly rates.
Rainfall ↔ Anomaly rate	0.599	Positive association suggesting that rainfall variability alone may not fully stabilise food prices due to lagged production responses and supply chain constraints.
SPI-3 ↔ Anomaly rate	-0.267	Moderate negative relationship showing that drought stress captured by SPI indicators is linked to increased food price spikes.
NDVI anomaly ↔ Mean price	-0.112	Weak negative relationship indicating that areas with poorer vegetation conditions tend to experience relatively higher food prices.
Rainfall ↔ Mean price	0.529	Moderate relationship reflecting spatial variation in agricultural productivity and climatic conditions across counties.
Price volatility ↔ Anomaly rate	0.095	Slight positive relationship indicating that markets with higher price volatility tend to experience more extreme price events.

3.2 Machine Learning Modelling

The modelling stage constituted the main component for this study, where supervised ML techniques were proposed for detecting anomalies in food commodity prices. The modelling process workflow began with feature engineering before proceeding to model design and training, and culminating in the classification of anomalous behaviour.

3.2.1 Feature engineering

In this study, feature engineering was carried out to construct explanatory variables that capture the complex drivers of food price behaviour. The process was undertaken to transform raw price, climate and macroeconomic variables into structured predictors suitable for machine learning models.

The engineered features focused on capturing temporal price dynamics, environmental conditions affecting agricultural production and broader economic influence on food markets. Table 6 summarises the variables that were used in the modelling process, including their categories, data sources and analytical rationale.

Table 6: Engineered Features Used in the Anomaly Detection Models

Feature	Category	Description and Source	Rationale
price_real	Price	CPI-deflated real commodity price (base year 2015=100), derived using WFP market prices and KNBS CPI.	Captures the inflation-adjusted price level used to analyse food price behaviour.
growth_rate	Price	Month-on-month percentage change in real price, derived from WFP price data.	Detects sudden price movements and forms the basis for identifying anomalies.
rolling_vol	Price	Six-month rolling standard deviation of the price growth rate.	Measures price instability; periods of high volatility often precede abnormal price spikes.
price_lag1	Temporal (lag)	Real price lagged by one month.	Captures short-term serial dependence in food commodity prices.
price_lag2	Temporal (lag)	Real price lagged by two months.	Represents medium-term momentum in price dynamics.
price_lag3	Temporal (lag)	Real price lagged by three months.	Captures quarterly patterns in price fluctuations.
price_roll3	Temporal (rolling)	Three-month rolling mean of real price.	Smooths short-term fluctuations and reduces noise in the price series.
NDVI_anomaly	Agro-ecological	NDVI deviation from county-month historical mean derived from MODIS MOD13Q1 data (Google Earth Engine).	Proxy for vegetation health; negative values indicate drought stress affecting crop yields.
NDVI_lag1	Agro-ecological (lag)	NDVI anomaly lagged by one month.	Captures delayed climatic effects on agricultural production and food markets.
rain_mm	Agro-ecological	Monthly cumulative rainfall derived from CHIRPS precipitation dataset.	Represents water availability for crops; low rainfall can lead to supply shortages and price spikes.
SPI3	Agro-ecological	Standardised Precipitation Index calculated over a 3-month window using CHIRPS rainfall data.	Indicates short-term drought conditions affecting agricultural output.
SPI6	Agro-ecological	Standardised Precipitation Index calculated over a 6-month window.	Captures medium-term drought stress influencing food availability.
SPI3_lag1	Agro-ecological (lag)	One-month lag of the SPI-3 drought index.	Provides an early warning signal of drought impacts on food prices.
fpma_maize	Inter-commodity	International maize reference price from the FAO FPMA dataset (Argentina export market).	Captures global price transmission affecting domestic food markets.
inflation_pct	Macroeconomic	Monthly inflation rate published by the Kenya National Bureau of Statistics (KNBS).	Reflects macroeconomic conditions influencing purchasing power and market prices.

Total engineered features: 15 Categories: 7 — Price, Temporal (lag), Temporal (rolling), Agro-ecological, Agro-ecological (lag), Inter-commodity, Macroeconomic Feature columns used in modelling: 15

The feature engineering was based on four dimensions of influence. First, the temporal features were generated through seasonal dummies, Fourier terms and lagged prices to represent harvest cycles and cyclical variation. Secondly, spatial features were introduced using neighbouring market prices, distance to Nairobi and road accessibility indicators to reflect trade networks and regional dependencies. Thirdly, agro-ecological features were derived from rainfall anomalies, the SPI and NDVI, providing signals of production shocks and drought stress. Finally, the inter-commodity features were created through price ratios and co-movement indices between staple crops and energy markets to capture cross-market linkages. These features

were meant to enrich the input space of the machine learning models, ensuring that anomaly detection reflects both statistical deviations and the underlying economic and ecological realities of Kenyan food markets.

Lastly, multicollinearity among the engineered variables was examined using a Pearson correlation matrix. Figure 3.9 presents the correlation matrix used to assess multicollinearity among the climate indicators and their lagged variables. The results showed a relatively strong correlation between each variable and its lagged counterparts, such as NDVI anomaly with NDVI_lag1 ($r = 0.75$) and SPI3_lag1 ($r = 0.68$), denoting the temporal persistence of climatic conditions. Moderate correlations were also observed between NDVI anomalies and SPI indicators, reflecting the influence of vegetation health and precipitation variability in capturing related but distinct aspects of agro-econological conditions.

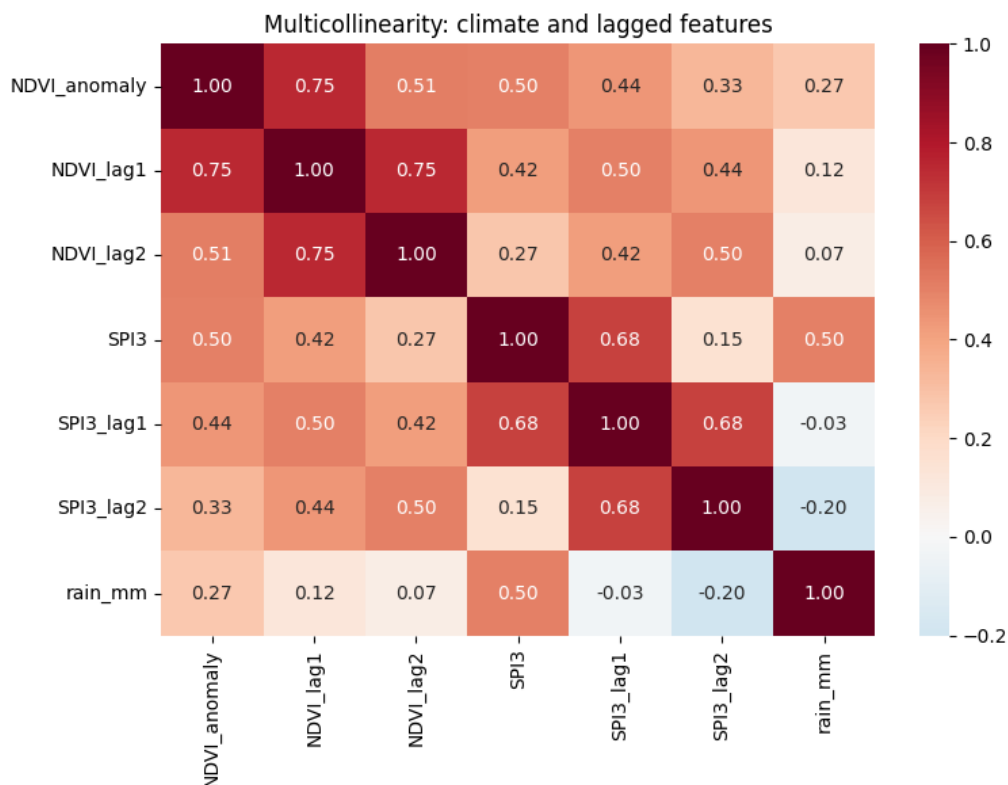


Figure 3.9: Correlation heatmap illustrating multicollinearity among climate and lagged variables. The matrix displays pairwise Pearson correlation coefficients between NDVI anomalies, rainfall, and standardised precipitation indices (SPI3), along with their lagged values.

Rainfall exhibited weaker correlations with most lagged variables, suggesting that it provides complementary information rather than redundant signals. Hence, there were no pairwise correlations approach levels naturally associated with severe multicollinearity, implying that the selected features significantly contribute distinct environmental signals that warrant their

inclusion in anomaly detection models.

3.2.2 Model Selection

The supervised anomaly detection task was formulated as a binary classification problem. This means that for each observation x_i , the objective is to predict a label $y_i \in \{0, 1\}$, with 1 denoting an anomaly and 0 denoting a normal observation. The central objective is to approximate a function $f : X \rightarrow Y$ that minimises a weighted loss function, $L(y, \hat{y})$ designed to handle the class imbalance inherent in anomaly detection

In this study, the model selection was guided by the need to balance predictive performance, interpretability and robustness in detecting food price anomalies. Five supervised machine learning algorithms were considered, that is, Linear Regression, the One Class Support Vector Machines (SVM), Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost) and Decision Trees.

3.2.3 Train - Test Strategy

Since food commodity prices follow a temporal structure, a time-based train-test split, that is, a rolling-origin evaluation strategy, was adopted to avoid information leakage. Unlike random sampling, time-based splitting ensures that models are trained only on historical observations and evaluated on future periods, which better reflects real-world forecasting conditions ([Hall and Rasheed, 2025](#)).

Before model training, the rows containing missing lagged price values were removed using *dropna*, while missing climate variables were imputed with zero to preserve temporal continuity. This preprocessing step helped reduce the anomaly-labelled observations from 139,043 to 235,699, ensuring that the final feature dataset retained valid lagged price information while maintaining consistent climate inputs.

The modelling dataset (*feat_df*), therefore, contained observations with full price and lagged features, while missing agro-economic variables were replaced with neutral values. The final dataset included 135,699 anomaly labelled observations.

A chronological split was then applied, where 2006 – 2022 as the training period and 2023 – 2025 was the testing period. Therefore, the training set contained 222,473 while the test set contained 36,394 observations. The proportion of anomalies remained relatively balanced across the sets, with an anomaly share of 52.83% in the training data and 49.95% in the test data, corresponding to 117,522 anomalies in the training period and 18,177 anomalies in the test period.

This rolling-origin evaluation framework ensured that the machine learning models are trained using only past information while being evaluated on future observations, providing a realistic assessment of their ability to detect emerging food price anomalies.

3.2.4 Baseline Model: Logistic Regression

Logistic Regression was used as a baseline classifier for anomaly detection because of its simplicity and interpretability in binary classification assignments. The model estimates the probability that a given observation represents an anomaly based on a linear combination of engineered features (Labhade-Kumar, 2024). The decision function is defined using a logistic (sigmoid) transformation shown in Equation 4

$$P(y = 1 | x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (4)$$

where x represents the feature vector, β denotes the vector of coefficients, and β_0 is the intercept term. The observations with predicted probabilities above a chosen threshold are classified as anomalies. In this study, logistic regression served as a benchmark for comparing the performance of more complex models in detecting abnormal food commodity price movements.

3.2.5 Support Vector Machine

A Linear SVM was preferred for their suitability in anomaly detection because they construct decision boundaries that separate normal and anomalous observations, even in high-dimensional spaces (Erfani et al., 2016). These models classify anomalies by finding a hyperplane that maximises the margin between normal and anomalous classes. The decision function is expressed as:

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right), \quad (5)$$

where α_i are the support vector weights, y_i are the class labels, $K(\cdot, \cdot)$ is the kernel function, and b is the bias term (Cortes and Vapnik, 1995). This formulation allowed SVMs to handle non-linear relationships common in food price data.

3.2.6 Decision Tree Model

Decision trees were included as a baseline model because of their interpretability and ability to capture non-linear relationships in food price dynamics. At each node, the tree selects a feature split that maximises the information gain by reducing impurity in the child nodes. Impurity is

often measured using entropy expressed in Equation 6:

$$H(S) = - \sum_{c=1}^C p_c \log_2(p_c), \quad (6)$$

where p_c is the proportion of observations in class c , and C is the number of classes. Equation 7 expresses that the best split is chosen to maximise the information gain:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v), \quad (7)$$

where $IG(S, A)$ is the information gain of splitting dataset S on attribute A , and S_v is the subset of S corresponding to value v .

Weighted Cross-Entropy Loss Since anomalies are rare, the loss function was adjusted to give higher weight to misclassified anomalies as illustrated in Equation 8:

$$L(y, \hat{y}) = -w_1 y \log(\hat{y}) - w_0 (1 - y) \log(1 - \hat{y}), \quad (8)$$

where $w_1 > w_0$ ensures that anomalies contribute more strongly to the optimisation objective.

3.2.7 Ensemble Models

Extreme Gradient Boosting Model. The XGBoost was preferred for modelling complex non-linear interactions among temporal, spatial and agro-ecological features more effectively than simpler classifiers. Its scalability and efficiency also make it well-suited for handling multi-market food price data, where anomalies are rare (Chen and Guestrin, 2016). The optimisation objective combines a regularised loss and penalty on model complexity as expressed in Equation 9 and Equation 10, respectively:

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_k \gamma \Omega(f_k), \quad (9)$$

with

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2, \quad (10)$$

where l is the differentiable loss function, such as logistic loss, f_k are individual trees, T is the number of leaves, γ controls tree complexity, λ regularises leaf weights, and w represents leaf

weights. This ensures a balance between model flexibility and overfitting control.

LightGBM. LightGBM was implemented as an ensemble gradient boosting model because of its efficiency and strong performance on large tabular datasets. The algorithm builds decision trees sequentially, where each new tree is trained to minimise the loss of the previous ensemble (Ke et al., 2017). The model prediction at iteration t is expressed as shown in Equation 11:

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i), \quad f_k \in \mathcal{F} \quad (11)$$

where f_k represents the k -th decision tree in the ensemble and $\mathcal{F}$ denotes the space of possible regression trees. The training objective is to minimise the regularised loss function as illustrated in Equation 12:

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^t \Omega(f_k) \quad (12)$$

where $L(\cdot)$ is the loss function measuring prediction error and $\Omega(f_k)$ is a regularisation term controlling model complexity. The lightGBM employs a leaf-wise tree growth strategy that improves predictive accuracy while maintaining computation efficiency. In this study, LightGBM was included as an additional ensemble model to compare its performance with the primary XGBoost model in detecting food commodity price anomalies.

3.2.8 Training Pipeline

All models were implemented using a consistent training pipeline to ensure comparability of the results. Engineered features were first prepared and split into training and testing sets using a time-based strategy. The selected machine learning models were then trained, after which predictions were generated and evaluated using classification performance metrics. Figure 3.10 shows a summary of the machine learning workflow adopted in this study.

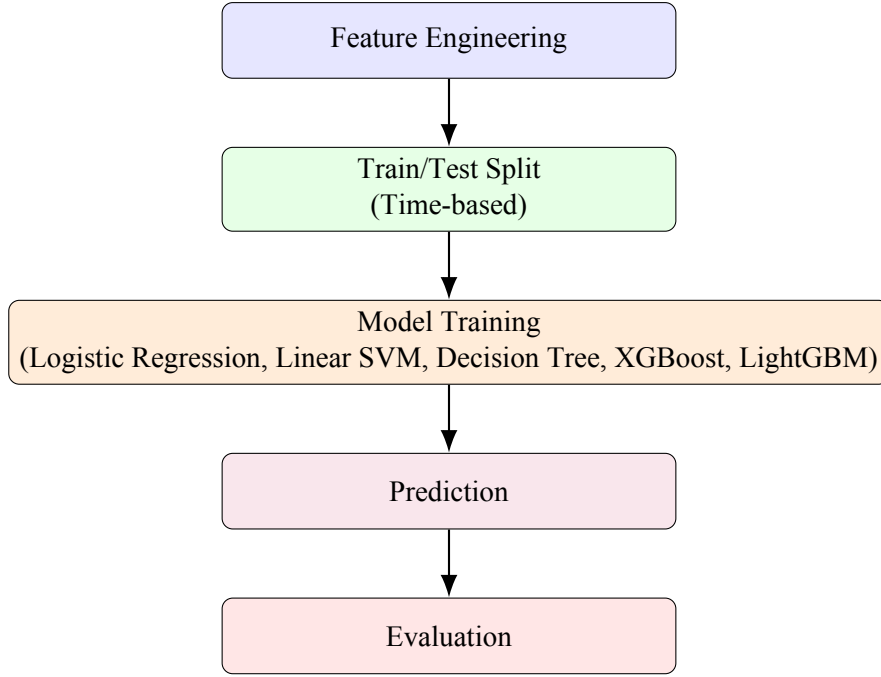


Figure 3.10: Model training pipeline used in the study.

3.2.9 Hyperparameter Tuning

This study conducted systematic hyperparameter tuning using a combination of cross-validation-based search and literature-informed parameter selection. The objective was to improve model performance while maintaining computational efficiency, given the large temporal dataset. Rather than conducting exhaustive grid searches across all algorithms, models were trained using predefined parameter configurations and evaluated using a time-based validation strategy (Sulandari et al., 2024).

Hyperparameters were specified using stable values commonly recommended for tabular machine learning tasks (Motz et al., 2022). These parameters control model complexity, learning rate and tree structure while limiting the risk of overfitting.

After model training, classification thresholds were calibrated using ROC analysis. Instead of adopting the default probability threshold of 0.5, the optimal threshold was selected using Youden’s J statistic, defined as shown in Equation 13:

$$J = \text{Sensitivity} + \text{Specificity} - 1 \quad (13)$$

where sensitivity represents the true positive rate and specificity represents the true negative rate. The threshold that maximises J provides the best balance between detecting anomalies and limiting false positives.

Using the ROC-optimised thresholds, the confusion matrix (see Figure 3.11) was generated for each of the five models to evaluate classification performance. This procedure ensured a fair comparison of model behaviour while accounting for the asymmetric costs of misclassifying food commodity price anomalies.

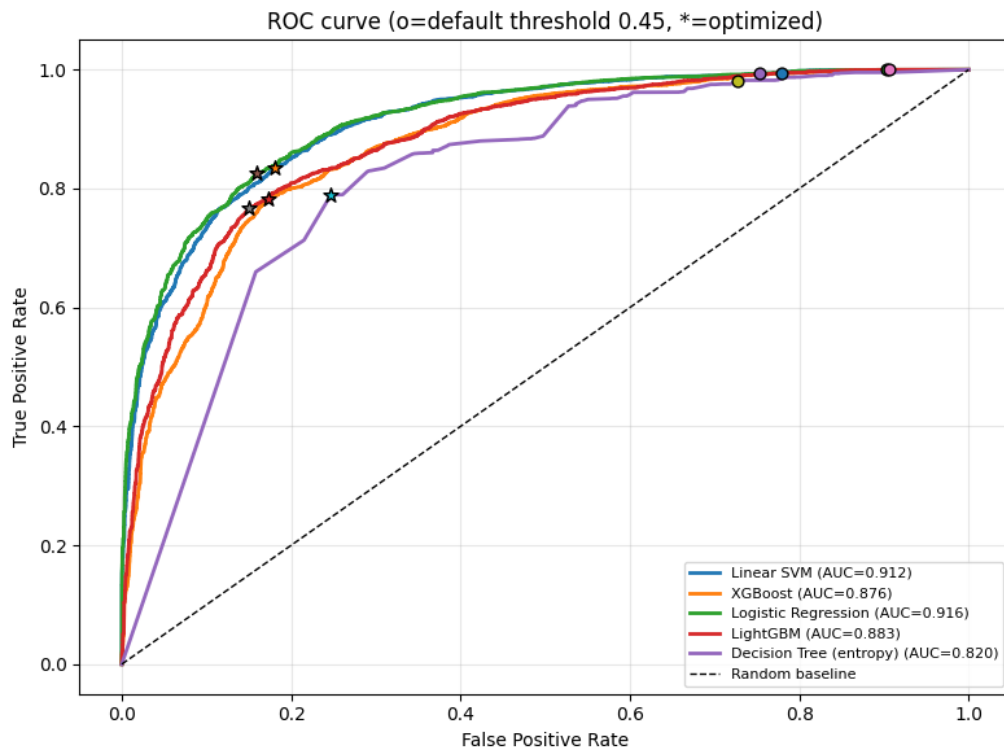


Figure 3.11: Comparative ROC curves for the evaluated anomaly detection models.

The results showed that the five models achieved good discrimination capability in identifying anomalous price behaviour. Logistic Regression produced the highest area under the curve (AUC = 0.916), closely followed by Linear SVM (AUC = 0.912), indicating that the linear models were particularly effective in separating normal and anomalous observations. Among the ensemble tree-based models, LightGBM achieved an AUC of 0.883, while XGBoost recorded a comparable performance with an AUC of 0.876. The Decision Tree model demonstrated relatively lower performance (AUC = 0.820), although it still maintained reasonable classification ability compared with a random baseline.

Lastly, overfitting was mitigated through several mechanisms, including the use of regularisation in linear models, depth constraints in the tree-based models and the use of a strict chronological train-test split. These measures ensured that the trained models generalised well to unseen future datasets.

3.3 Performance Evaluation

The model's performance was evaluated using a stratified 5-fold cross-validation design to test robustness across time and markets. The main metrics were defined as shown in Table 7.

Table 7: Performance evaluation metrics used in the study

Metric	Description	Mathematical Definition
Precision	Correctly identified anomalies among all predicted anomalies	$\text{Precision} = \frac{TP}{TP+FP}$
Recall	Actual anomalies correctly identified	$\text{Recall} = \frac{TP}{TP+FN}$
F1-score	Harmonic mean of precision and recall	$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
True Positive Rate	Proportion of anomalies detected	$TPR = \frac{TP}{TP+FN}$
False Positive Rate	Proportion of normal cases incorrectly flagged	$FPR = \frac{FP}{FP+TN}$
Area Under the Curve	Overall ability of the model to separate normal from anomalous cases	$AUC = \int_0^1 TPR(FPR) d(FPR)$

Lastly, the confusion-matrix analysis complemented these metrics to show trade-offs between false alarms and missed anomalies. Because missed anomalies carry greater policy risk than false positives, recall received greater weight in the final model selection.

3.4 Model Deployment

Figure 3.12 is an illustration of the deployment process for the final model, starting with data input to and market use.

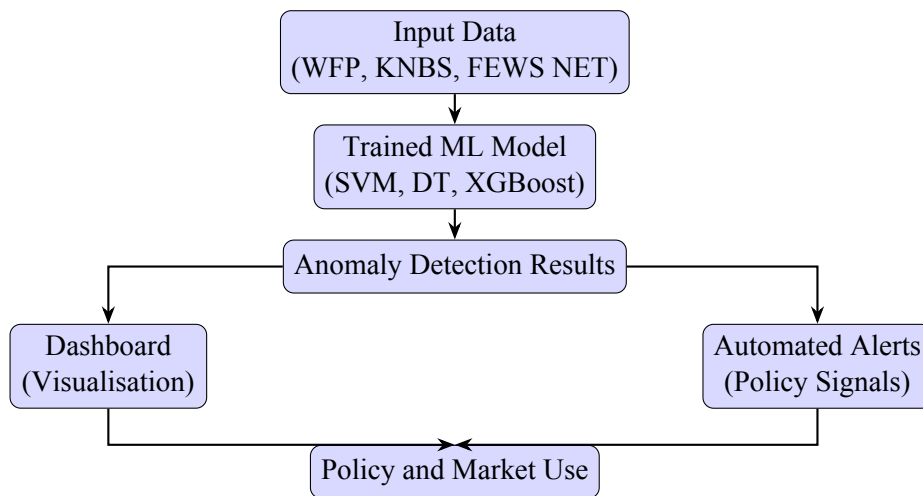


Figure 3.12: Deployment workflow for anomaly detection outputs integrated into dashboards and alerts.

The model was deployed as a cloud-based system that connects directly to market data

from WFP, KNBS, and FEWS NET. New data was processed in real time by the trained algorithms to detect anomalies and produce price forecasts. Results feed into two tools: a dashboard that visualises market patterns and anomalies against historical trends, and an alert system that issues early warnings on unusual changes in staple food prices such as maize, beans, and rice. These outputs supports farmers, traders, and policymakers by providing timely insights for both immediate response and long-term planning. The use of cloud infrastructure was meant ensure fast processing, secure access, and scalability, while automated monitoring and continuous learning keep the models updated and adaptable to changing market and climate conditions.

3.5 Expected Outcomes

This study provide a supervised machine learning framework that identifies food price anomalies in near real time using routinely collected market data. The model brings seasonality, spatial differences across Kenyan markets and links between commodities into one pipeline to lift detection accuracy over standard baselines. The workflow was packaged so it can be ported to additional markets with minimal reconfiguration, allowing replication in neighbouring countries once local data is available.

The practical output was intended to be decision-ready signals for analysts and policymakers. The system generates weekly anomaly alerts, summary briefs and simple visual explanations that show why a flag was raised and how current prices compare with seasonal norms. These outputs are meant to strengthen routine market monitoring, support targeted policy actions such as temporary subsidies or import timing and inform humanitarian planning before sharp price swings spill over into food insecurity. By shortening the time from data ingestion to a clear alert, the project aims to improve early warning agencies that track market stability.

CHAPTER 4: SYSTEM DESIGN AND ARCHITECTURE

This chapter presents the design and architecture of the anomaly detection system developed to identify irregular patterns in food commodity prices in Kenya. The system architecture defines the structural components of the solution and explains how data, analytical models and user interfaces interact to support the detection of anomalous price movements. The aim is to organise the system into clearly defined modules to ensure that the architecture can effectively collect, process, analyse and visualise food price data to provide timely anomaly alerts.

The architecture was designed to support the machine learning workflow used for anomaly detection. It integrates data ingestion, preprocessing, feature engineering, model execution, and visualisation components into a unified pipeline. Through this pipeline, historical and incoming market data are transformed into analytical features and processed by supervised machine learning models to detect deviations from expected price behaviour. The resulting outputs are presented through dashboards and alerts that support market monitoring and early warning.

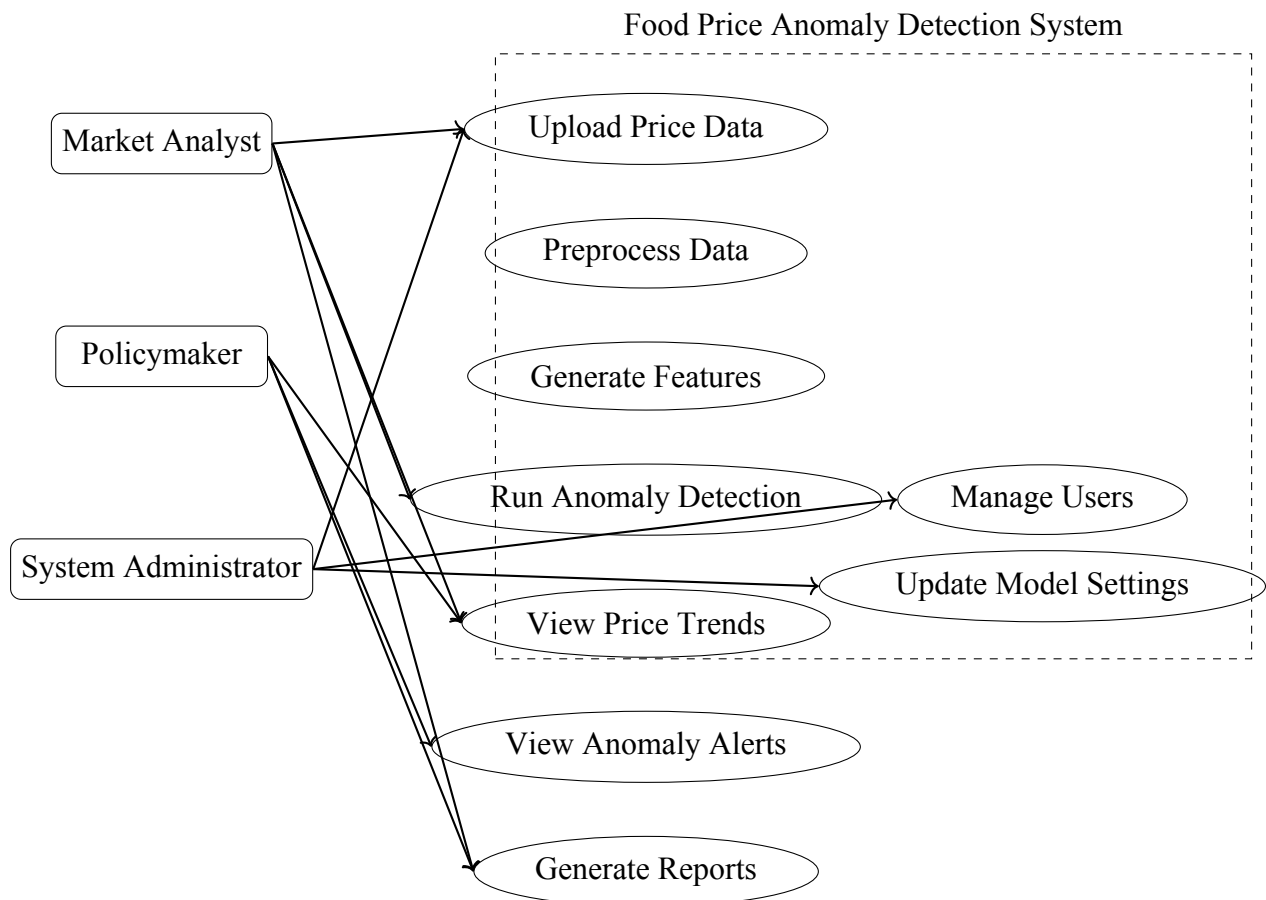


Figure 4.1: System Use Case Diagram for the Food Price Anomaly Detection System

Figure 4.1 is an illustration of the interactions between users and the anomaly detection

system. The market analysts interact with the system to upload and analyse price data, policymakers access anomaly alerts and reports for decision-making, while system administrators manage user access and maintain the system configuration.

The system design directly reflects the methodological framework described in Chapter 3, which follows the CRISP-DM process for data mining. Each architectural component corresponds to a stage of the methodology, including data understanding, data preparation, modelling, evaluation and deployment. This alignment ensures that the implementation of the system follows a structured and reproducible analytical process, allowing the developed framework to be adapted to other markets or datasets in future applications.

4.1 System Requirements Analysis

4.1.1 Functional Requirements

Figure 4.2 presents the functional requirements architecture of the food price anomaly detection system. The architecture organises the functional requirements of the system into five core modules.

The system includes a data ingestion component that allows food commodity price datasets from sources such as the WFP, the KNBS and FEWS NET to be loaded into the analytical environment. A data preprocessing component was implemented to clean and prepare the datasets. This includes handling missing values, adjusting price values using the consumer price index and transforming the data into a consistent format suitable for analysis.

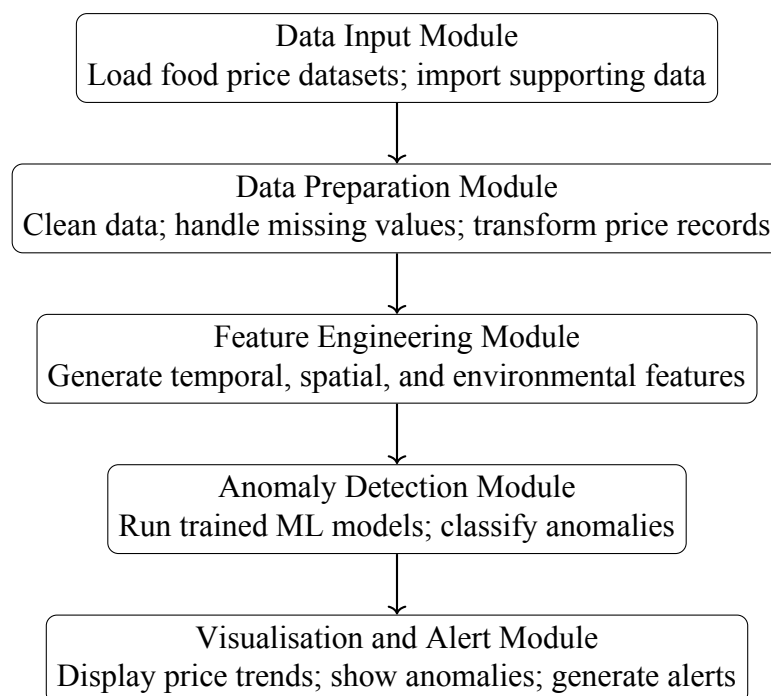


Figure 4.2: Functional requirements architecture of the food price anomaly detection system

The system also includes a feature engineering module that generates explanatory variables representing temporal patterns, spatial relationships between markets, agro-ecological indicators and inter-commodity price relationships. Equally, an anomaly detection module was incorporated to execute the trained machine learning models developed in this study. The models analyse the engineered features and classify observations as either normal price behaviour or anomalous price movements.

Finally, the system includes a visualisation and alert component that displays price trends and detected anomalies through a dashboard interface. This component also generates anomaly alerts to highlight unusual price movements in specific commodities and markets.

4.1.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes that support the effective operation of the anomaly detection system. These requirements ensure that the system performs reliably and efficiently while supporting the analytical tasks defined in the study.

The system was designed to support scalability by enabling the preprocessing of datasets across multiple markets and commodities. Performance considerations ensure that data processing and model execution are carried out efficiently to enable timely anomaly detection. The issue of usability was addressed by implementing a dashboard that lets users easily visualise price trends and detect anomalies.

Further, reliability was incorporated by ensuring that the system can handle incomplete or inconsistent datasets through preprocessing techniques such as interpolation and data cleaning. Security considerations were included to control access to the system and protect the integrity of the data and analytical outputs. Finally, maintainability was supported by a modular system architecture that allows individual components to be updated or improved without affecting the overall system functionality.

4.2 Data Architecture Design

This section describes how data used in the anomaly detection system is collected, stored and prepared for machine learning analysis. The architecture defines the flow of data from external sources to the analytical environment where preprocessing, feature engineering and modelling are performed.

4.2.1 Data Sources

The system integrates multiple datasets to support anomaly detection in food commodity prices. The primary data source consists of monthly retail and wholesale price observations for staple

commodities such as maize, beans and rice obtained from the WFP Price Database through the OCHA Services platform.

Additional supporting datasets are incorporated to improve the analytical context of the model. Consumer Price Index and official retail statistics from the KNBS are used to adjust nominal prices and maintain consistency with national economic indicators. International reference prices from the FPMA tool are used for comparative analysis of domestic price movements, while FEWS NET market bulletins provide contextual information on market shocks such as droughts, supply disruptions and trade restrictions.

Auxiliary datasets are also integrated to capture environmental and spatial drivers of food price behaviour. These include rainfall anomalies, the SPI and the NDVI for climatic variability, as well as infrastructure indicators such as road density, distance to Nairobi and proximity to trade corridors. These variables help capture the agro-ecological and spatial factors that influence price dynamics across markets.

4.2.2 Data Architecture

The system stores and manages datasets through a structured data pipeline that separates raw data from processed analytical datasets. The data collected from different external sources are first stored in a raw data repository, where the original datasets are archived without modification. This layer preserves the integrity of the source data and allows reproducibility of the analysis. Figure 4.3 illustrates the data storage architecture showing the flow of datasets from external data sources to the model input dataset used for anomaly detection.

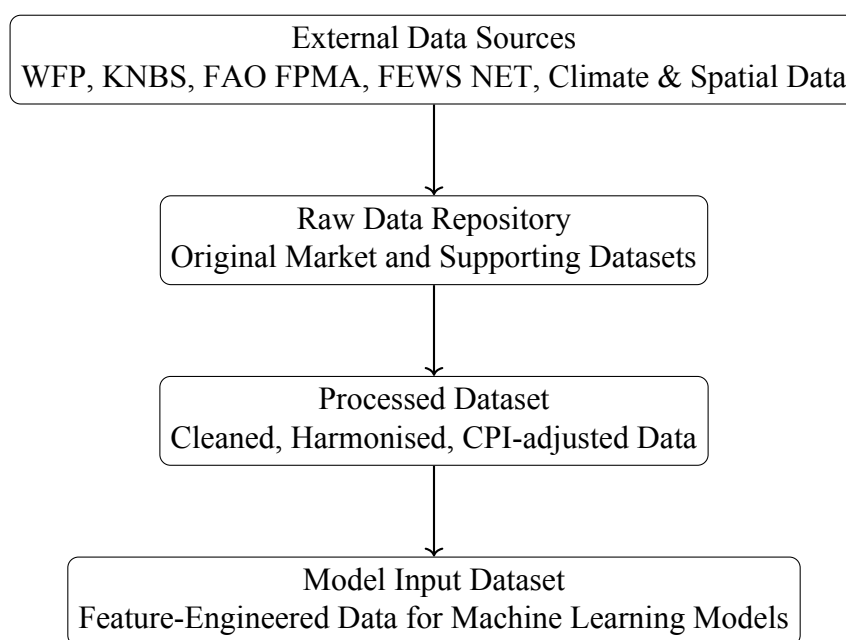


Figure 4.3: Data storage architecture of the food price anomaly detection system

Once ingested, the datasets are then processed through the preprocessing pipeline described in Figure 4.4, where missing values are handled, prices are adjusted for inflation using CPI, and data transformations are applied. The resulting cleaned data are ready for analytical use. The processed datasets are finally transformed through feature engineering procedures to generate model-ready inputs. These model input datasets contain the temporal, spatial, agro-ecological and inter-commodity features used by the machine learning models for anomaly detection.

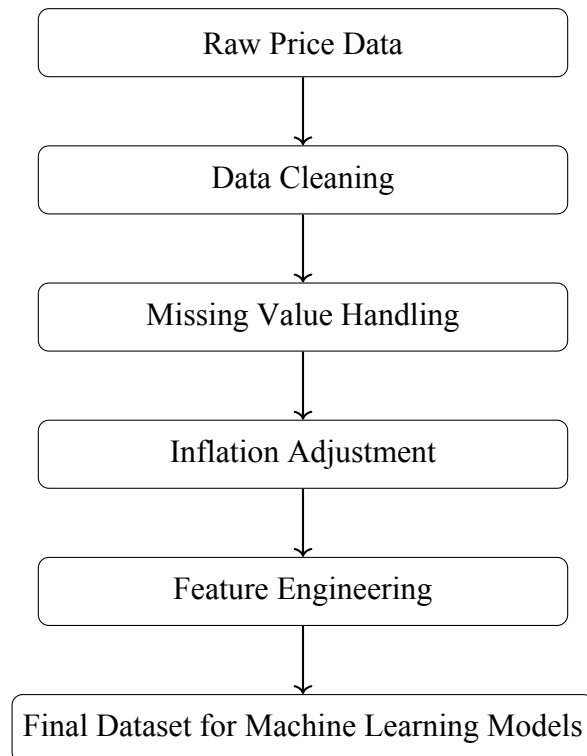


Figure 4.4: Data processing pipeline for the food price anomaly detection system

4.3 Machine Learning Architecture Design

4.3.1 Feature Engineering Architecture

The feature engineering module forms part of the data processing pipeline and is responsible for transforming the cleaned datasets into structured variables that can be used by the machine learning models.

Within the system, the feature engineering components receive the processed dataset generated by the preprocessing module. The system automatically derives several categories of features that represent different aspects of food price behaviour. Temporal features are generated from the price time series, including lagged price variables and growth rates. Spatial features are produced by incorporating price information from neighbouring markets and geo-

graphic indicators such as distance to major market centres. Climate-related features are derived from auxiliary datasets, including rainfall anomalies, the SPI and NDVI.

The system also constructs inter-commodity features that capture relationships between staple commodities. These features are generated using price ratios and co-movement indicators that reflect shared market dynamics. Once generated, all features are combined into a unified feature matrix that serves as the input dataset for the machine learning models responsible for anomaly detection. Figure 4.5 shows the feature engineering framework used to transform raw data into structured features for the machine learning models.

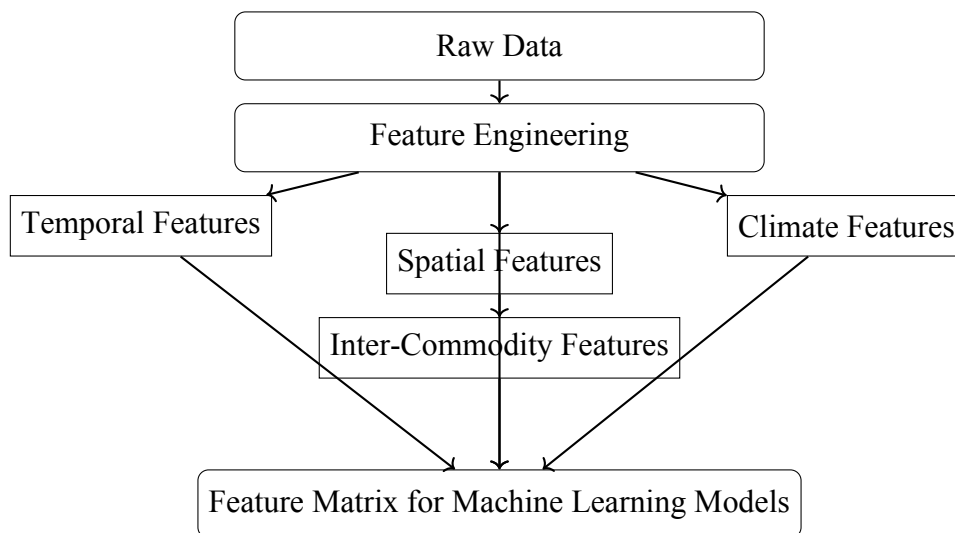


Figure 4.5: Feature engineering framework showing the categories of features used in the anomaly detection models

4.3.2 Model Architecture

The model architecture defines how engineered features are processed by the models to identify anomalous food price behaviour. The pipeline is organised into four components, that is, input features, model training, model evaluation and prediction layer as illustrated in Figure 4.6.

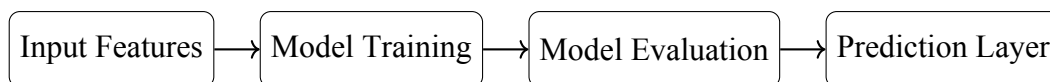


Figure 4.6: Machine learning pipeline of the anomaly detection system

The input features are obtained from the feature engineering module and structured into a feature matrix, forming an input to the machine learning models. During the training stage, the algorithms used in the study were trained using historical data to learn patterns associated with normal and anomalous price movements. After training, the models were evaluated using classification metrics, where the best model was deployed in the prediction layer, which processes

new observations and classifies them as normal or anomalous. These predictions are subsequently forwarded to the analytics and visualisation components of the system for monitoring and alert generation.

4.3.3 Model Management

Model management in the system was focused on the storage and reuse of trained models during the anomaly detection process. After training, the machine learning models developed in this study were serialised and stored using the Joblib library in Python 3.11.9. Joblib enables efficient saving and loading of trained models, allowing the system to extract the stored model directly during the prediction stage without repeating the training process. This mechanism improves computation efficiency and ensures that the trained models can be consistently reused for analysing new food price data.

The stored models were retained together with their configuration settings and evaluation results to support traceability and comparison during experimentation. When updated food price data becomes available, the training pipeline can be executed again to retrain the models and generate updated serialised versions. This process allows the system to maintain reliable anomaly detection performance while adapting to evolving market conditions.

4.4 User Interface and Dashboard Design

The anomaly detection system is implemented as a multi-page Streamlit web application accessed through a web browser. The interface is structured into three main components, that is, the sidebar control panel, main display area and page navigation. The sidebar allows users to select filters such as commodity, county, data range and severity level, while the main display area presents the analytical outputs, including key indicators, charts, maps and anomaly tables. The navigation links in the sidebar provide access to the three main pages of the application that are the Dashboard, Anomaly Detection and Interpretability.

Figure 4.7 shows the user interface for the Dashboard interface wireframe. User interaction within the system is primarily filter-driven and action-driven. On the Dashboard page, users apply filters to explore historical price trends and detect anomalies across commodities and counties. The Dashboard provides the main monitoring interface for anomalies by displaying key performance indicators such as commodities monitored, markets covered, anomalies detected in the latest month and average risk score. Additional visualisations include price trend charts with anomaly markers, severity distribution charts, anomaly trend over time, a county-level market risk map and a table of detected anomalies containing information such as

commodity, location, observed price, expected price, anomaly score and severity level.

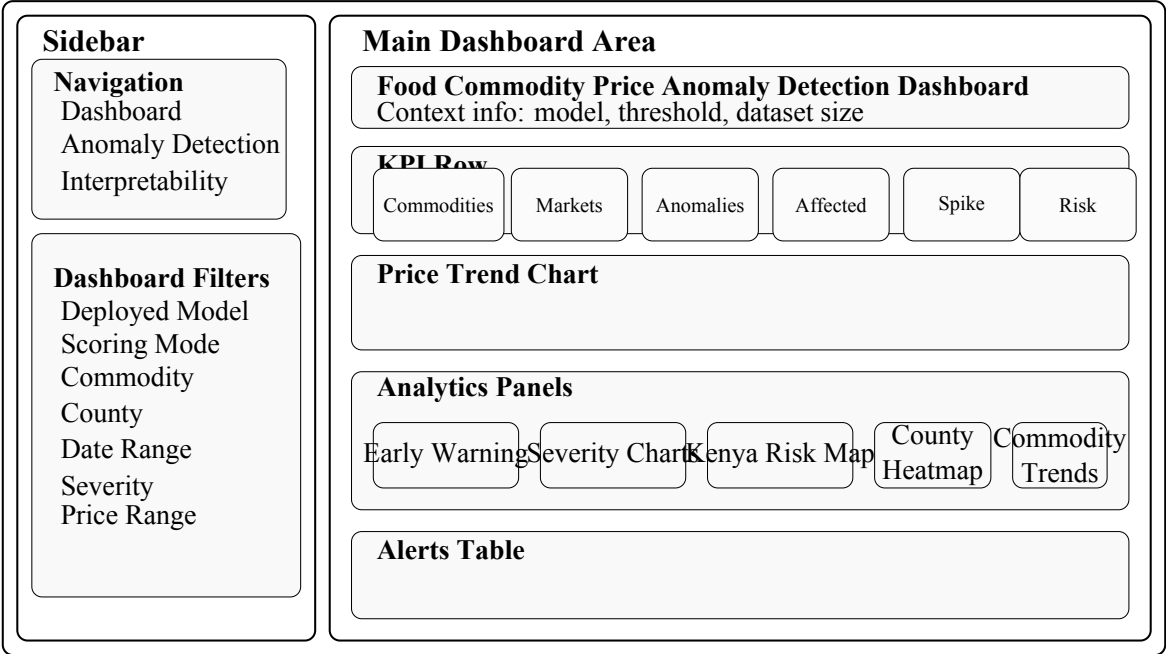


Figure 4.7: Dashboard wireframe showing the main interface sections of the anomaly detection system

The Anomaly Detection page enables users to upload datasets and execute the anomaly detection workflow. As illustrated Figure 4.8, the interface is divided into a sidebar ingestion panel and main analytical display area.

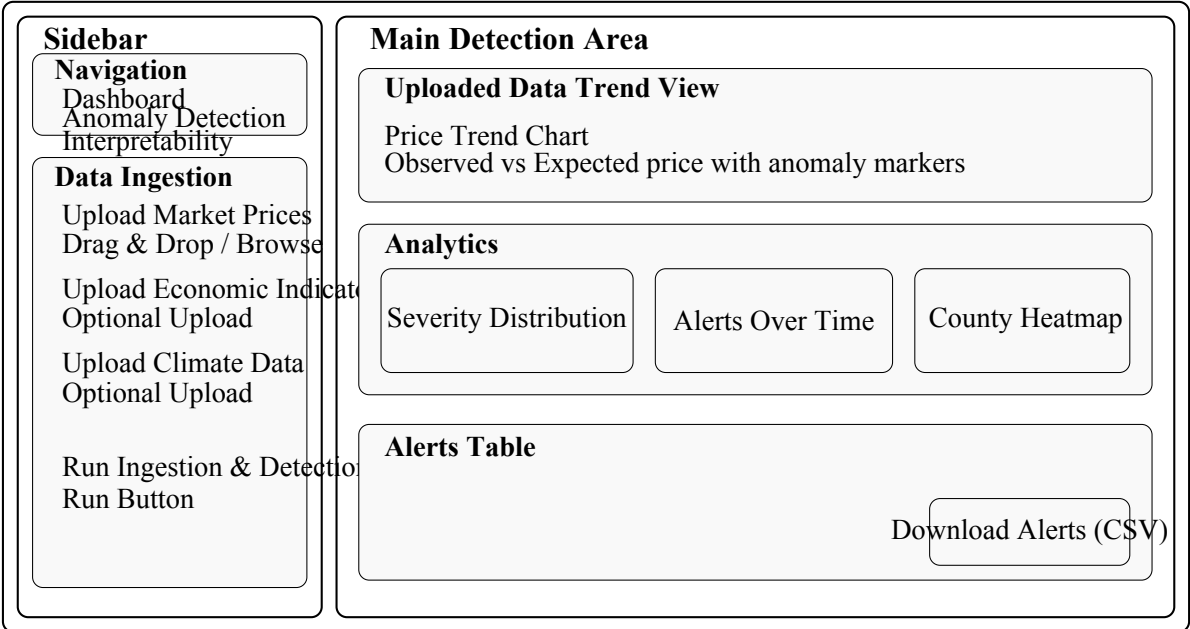


Figure 4.8: Wireframe of the anomaly detection page showing the data ingestion controls and analytical outputs

In the sidebar, users upload the required market price dataset and optional supporting datasets such as economic, climate or spatial data, then initiate processing through the Run In-

gestion and Detection action. The system validates and harmonises the data, applies the trained anomaly detection model and displays the results in the main area through a price trend chart with anomaly markers, severity and temporal charts, a county heatmap and an alerts table listing detected anomalies. Subsequently, a CSV download option allows users to export the alert results for further analysis.

The Interpretability page provides model explanation capabilities for anomaly predictions. As shown in Figure 4.9, users select filters such as commodity, county, model and threshold from the sidebar. The system then allows users to generate global SHAP summaries, which display feature importance based on mean SHAP values, and local SHAP explanations, which provide detailed interpretation for selected high-risk observations. This interface enables users to understand how different features contribute to the predictions by the anomaly detection model.

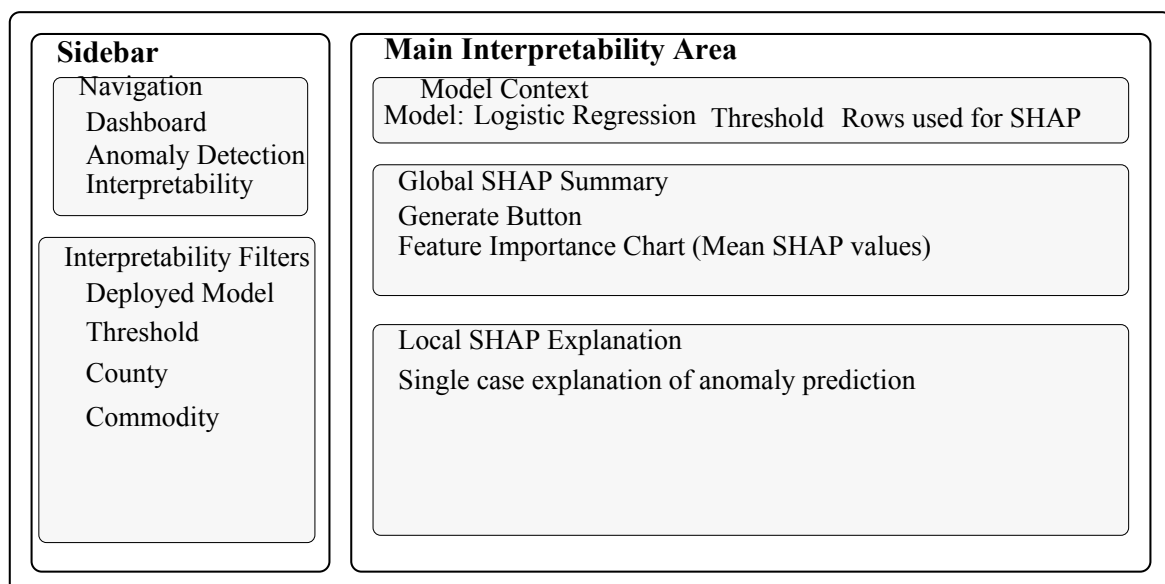


Figure 4.9: Wireframe of the interpretability page showing the SHAP explanation interface

4.5 Technology Stack

The system was developed using the Python ecosystem and integrates tools for data processing, machine learning, visualisation and web application deployment.

4.5.1 Programming Language and Runtime

The system is implemented in Python 3 (3.11.9), which served as the primary programming language for data processing, machine learning inference and the web application. Python was selected due to its extensive ecosystem of data science libraries and strong support for machine learning and web-based analytics applications.

4.5.2 Web Application Framework

The user interface of the system is implemented using Streamlit, which provides a lightweight framework for building interactive data applications. Streamlit was preferred because it supports the development of a multi-page application consisting of the Dashboard, Anomaly Detection and Interpretability pages. The framework enables the use of sidebar filters, file upload components, interactive charts and data tables. Streamlit caching mechanisms are used to optimise performance when loading data and model artefacts.

4.5.3 Data Processing and Preprocessing

Data manipulation and preprocessing operations were implemented using Pandas and NumPy. Pandas was used for loading datasets, performing time-series transformations, merging multiple data sources and generating feature tables. NumPy, on the other hand, supports the numerical operations used in preprocessing and anomaly scoring. The missing values, particularly in monthly price series, are handled using the KNNImputer provided by scikit-learn, while openpyxl enables ingestion of uploaded Excel datasets.

4.5.4 Machine Learning and Model Serving

Machine learning components are implemented using scikit-learn and XGBoost. The models evaluated include Logistic Regression, Linear Support Vector Machine, Decision Tree, LightGBM and XGBoost, with Logistic Regression selected as the final best deployed model based on its superior classification performance. Model training and evaluation were performed in the development notebook environment, after which the selected model was integrated into the application for anomaly detection. For deployment, the trained model and associated preprocessing artefacts are stored as serialised objects using the joblib library. These artefacts include the trained Logistic Regression, feature scaler, and anomaly detection threshold, which are saved in the system output directory.

4.5.5 Model Explainability

Model interpretability is implemented using the SHapley Additive exPlanations (SHAP) library. SHAP provides both global and local explanations of the model predictions. The LinearExplainer is used for interpreting linear models such as Logistic Regression, while TreeExplainer is used when analysing tree-based models. These explanations are displayed on the Interpretability page of the application.

4.5.6 Visualisation

Interactive visualisations are implemented using Plotly, which supports dynamic charts within the Streamlit interface. Plotly is used to generate time-series charts for price trends, bar charts for anomaly severity and feature importance, line charts for anomaly trends over time, heatmaps for county-level anomaly patterns and geographic scatter plots for the Kenya market risk map.

4.5.7 Data Sources and Storage

The system uses multiple CSV datasets as inputs, including market price data, inflation data and international reference prices. The model artefacts and configuration files are stored within the project directory in formats such as CSV, JSON and Joblib. The application does not rely on a separate database; instead, data are loaded from local files and user uploads, with intermediate results stored in memory using Streamlit caching.

4.5.8 Deployment and Execution

The application runs using the Streamlit runtime environment, which enables execution of the dashboard using the command `streamlit run app.py`. Configuration settings for the server and visual theme are defined in the `.streamlit/config.toml` file. The system can be executed locally or deployed on supported hosting environments such as Streamlit Community Cloud.

Table 8 is a summary of a combination of Python-based data processing, machine learning, and visualisation technologies that were used in the system implementation.

Table 8: Summary of Technology stack used in the anomaly detection system

Layer		Technology	Role
Programming Language	Lan-	Python 3	System implementation
Web Framework		Streamlit	Dashboard interface and application pages
Data Processing		Pandas, NumPy	Data manipulation and feature generation
Preprocessing		scikit-learn, openpyxl	Missing value imputation and Excel ingestion
Machine Learning		scikit-learn, XGBoost	Model training and prediction
Model Storage		Joblib	Serialization and loading of trained models
Explainability		SHAP	Global and local model explanations
Visualisation		Plotly	Interactive charts and maps
Data Storage		CSV, JSON, Joblib	Input datasets and model artifacts
Deployment		Streamlit runtime	Application execution

CHAPTER 5: SYSTEM IMPLEMENTATION AND TESTING

This chapter presents the implementation and testing of the food price anomaly detection system developed in this study. Its purpose is to show how the architectural design described in Chapter Four was translated into a functional application for detecting anomalous food price behaviour. The chapter, therefore, focuses on the system’s operational aspects, including the implemented modules, the user interface, and the analytical components.

5.1 System Implementation

The anomaly detection system is implemented as a hybrid architecture combining Streamlit and FastAPI, where Streamlit provides the interactive user interface, and FastAPI serves as the backend for scalable model inference. The Streamlit application is structured into multiple pages, including the Dashboard, Anomaly Detection and Interpretability modules, enabling users to perform monitoring, data ingestion and model explanation within a unified interface. User interactions such as file uploads, filter selection and visual exploration are handled within Streamlit, while computationally intensive tasks are delegated to the backend service.

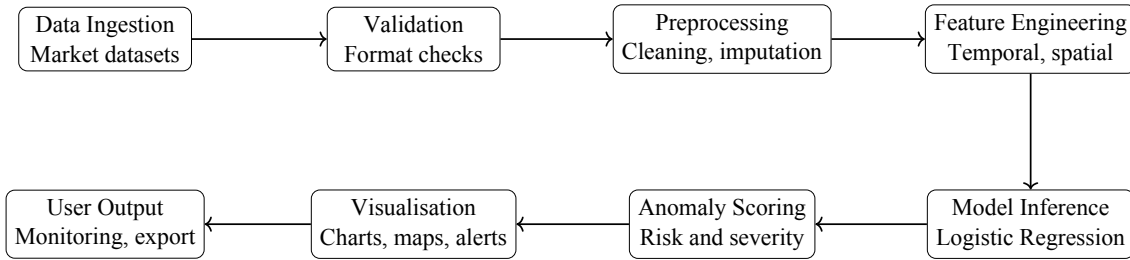


Figure 5.1: System implementation workflow showing data flow from ingestion to dashboard outputs

The FastAPI layer is introduced to handle data processing and model inference in a scalable and modular manner. When users upload datasets or trigger anomaly detection, the Streamlit frontend sends requests to FastAPI endpoints, where the data pipeline is executed. This pipeline includes data validation, preprocessing, missing value handling and feature engineering before passing the processed data to the deployed model. This separation of concerns improves system performance, supports scalability and enables the backend to be extended for future integration with external systems or APIs. Figure 5.2 shows the interface used to enable remote scoring through the FastAPI backend.

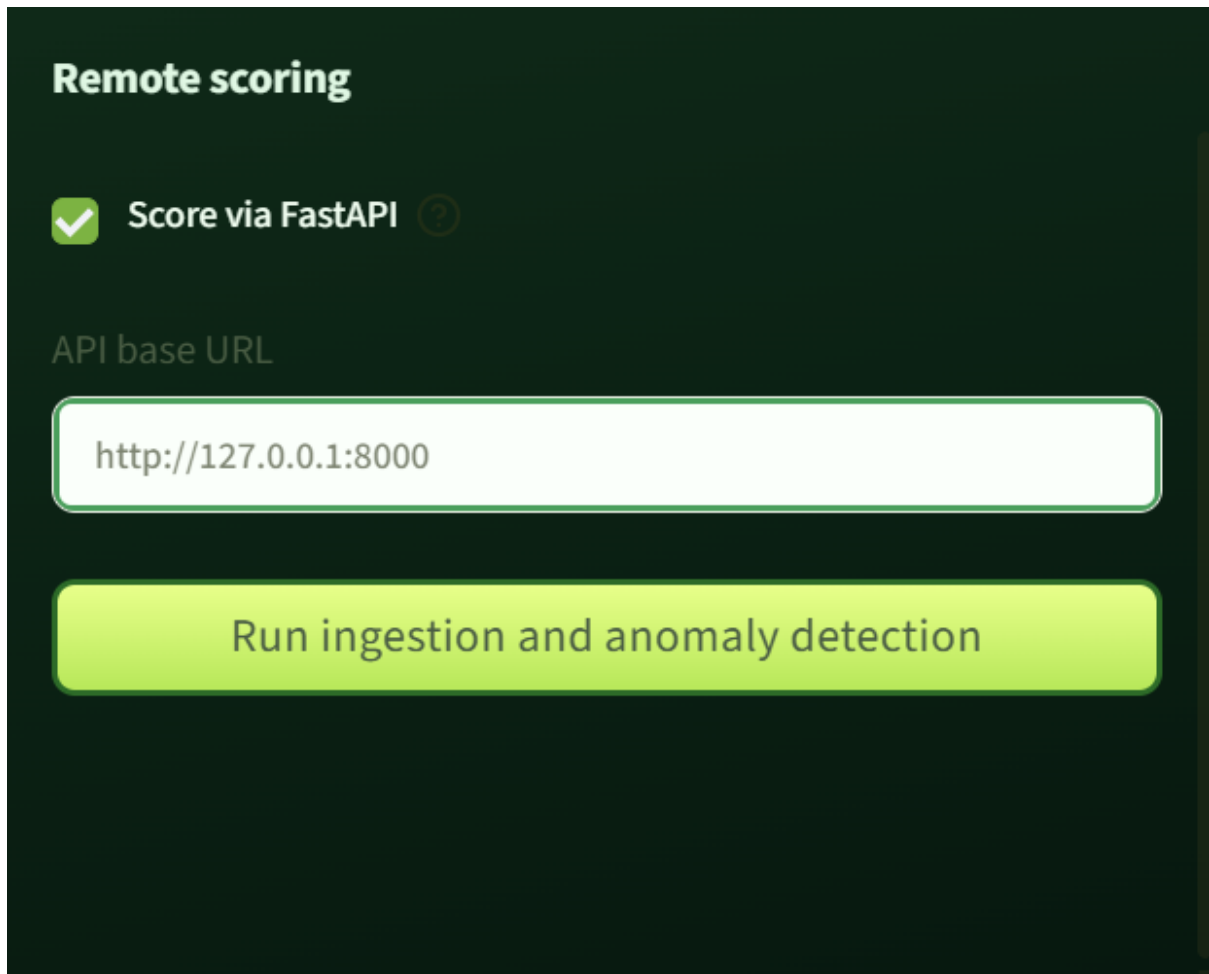


Figure 5.2: FastAPI integration interface used for remote anomaly scoring

The Logistic Regression model, selected based on performance evaluation, is deployed within the FastAPI service as the core anomaly detection component. Model artefacts, including the trained model, feature scaler and the anomaly detection threshold, are stored as serialised objects and loaded using the joblib library during API initialisation. Upon receiving a request, the FastAPI service applies these artefacts to the processed datasets to generate anomaly predictions, which are then returned to the Streamlit interface for visualisation through charts, maps and alert tables. This architecture ensures efficient model reuse, real-time prediction and improved scalability of the anomaly detection system.

5.2 System Interface Demonstration

5.2.1 Dashboard Interface

The Dashboard interface provides the primary environment for monitoring food price anomalies across commodities and regions. The interface displays key indicators, trend visualisations, geographic risk patterns and anomaly alerts, enabling users to analyse price behaviour in real

time. Figure 5.3 presents the dashboard interface, illustrating the sidebar filters, model configuration and interactive controls used to monitor food price anomalies across commodities and counties.

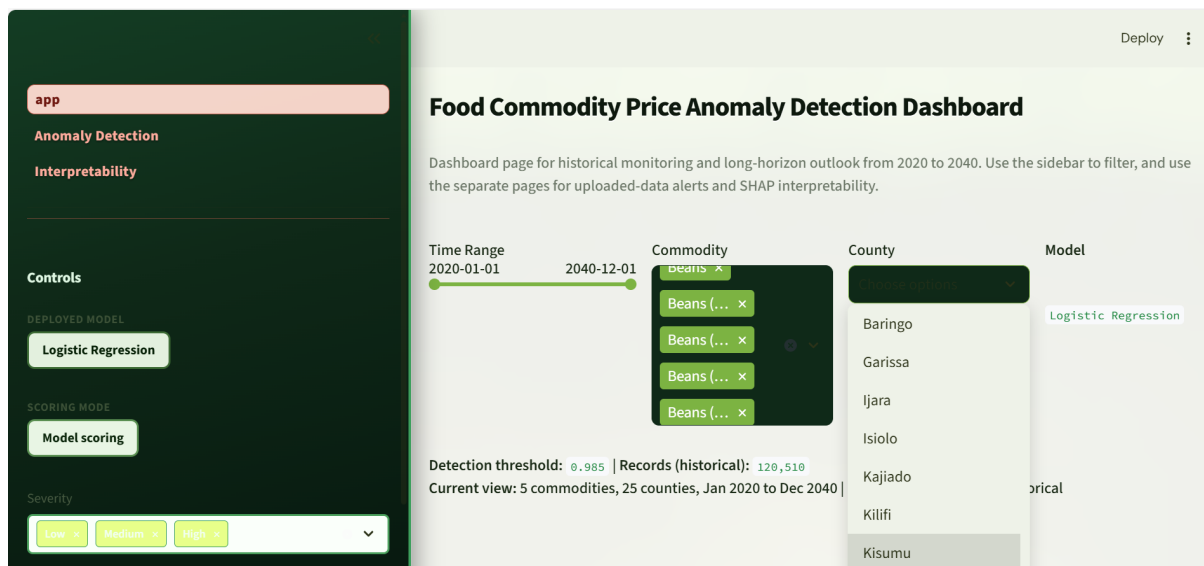


Figure 5.3: Dashboard interface showing sidebar filters, deployed model settings, and monitoring controls for anomaly detection

The dashboard includes a Key Performance Indicator (KPI) section that summarises key metrics such as the number of commodities monitored, markets covered, detected anomalies, the most affected commodity, the highest price spike and average risk score. These indicators provide a quick overview of market conditions. Figure 5.4 presents key dashboard analytical components, including summary indicators, anomaly visualisations, and statistical insights used for monitoring food price behaviour.

The price trend chart presents observed prices alongside expected values, with anomaly markers indicating abnormal price movements. This allows users to identify deviations indicating abnormal price behaviour over time. In addition, the Kenyan market risk map visualises the spatial distribution of anomalies, highlighting counties with elevated anomaly levels.

The anomaly alerts table (panel) in Figure 5.5 provides detailed information on detected anomalies, including date, commodity, county, observed and expected prices, anomaly score, and severity level. Users can use this table to investigate specific anomaly events.

User interaction is controlled through sidebar filters, which allow selection of commodity, county, date range, severity level and price range. These filters dynamically update all dashboard components, enabling users to explore anomalies across different dimensions.

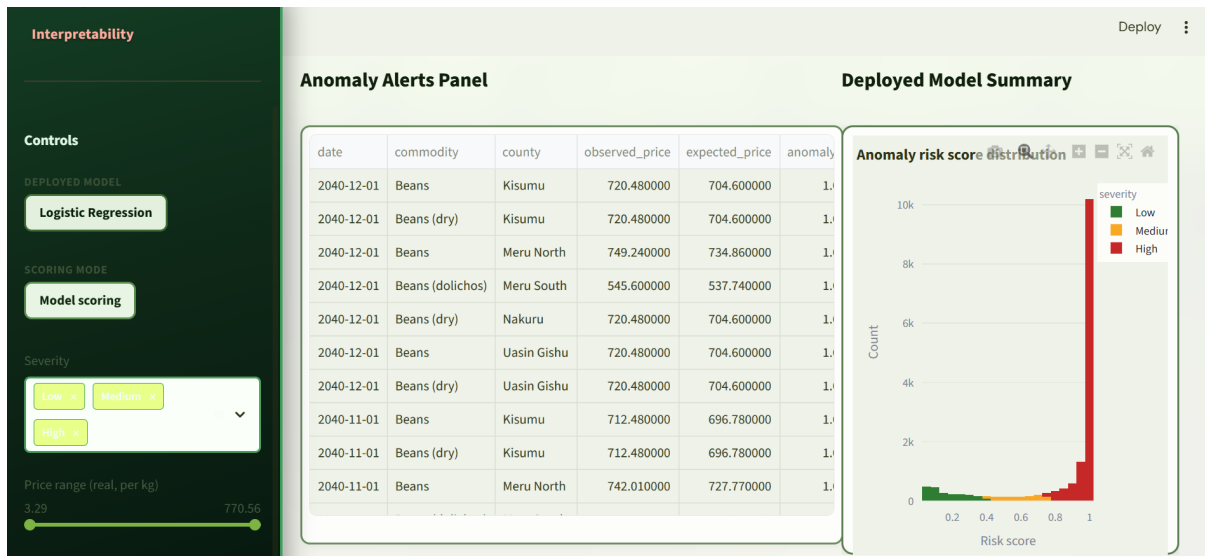


Figure 5.5: Anomaly alerts panel displaying detected anomalies and deployed model summary showing risk score distribution across severity levels

5.2.2 Anomaly Detection Interface

The Anomaly Detection Interface enables users to upload datasets and execute the anomaly detection pipeline. The interface consists of a data ingestion panel and a results display section. Figure 5.6 illustrates the anomaly detection interface, including FastAPI-based remote scoring and the multi-source data ingestion components used for model execution.

Users upload the required dataset and optional supporting datasets, such as economic indicators or climate data, through the sidebar. Once the datasets are provided, the user initiates the process by selecting the **Run ingestion and anomaly detection** option. The system then validates and processes the data, performs feature engineering and applies the deployed Logistic Regression model to generate anomaly predictions.

As shown in Figure 6.2, the results are displayed through a price trend chart with anomaly markers, allowing users to visualise how anomalies occur within the uploaded dataset. Additional visualisations include severity distribution charts, alerts over time, and a county heatmap, which provide insights into the distribution and patterns of anomalies.

The interface also presents an anomaly alert table, which lists all detected anomalies with their associated attributes. A download option is provided to allow users to export the results in CSV format for further analysis.

<<

app

Anomaly Detection

Interpretability

Data Ingestion

Upload Market Prices

Drag and drop file here
Limit 200MB per file • CSV, XLSX

Browse files

Upload Economic Indicators

Drag and drop file here
Limit 200MB per file • CSV, XLSX

Browse files

Remote scoring

☒ Score via FastAPI

API base URL

http://127.0.0.1:8000

Run ingestion and anomaly detection

Figure 5.6: Anomaly detection interface showing dataset upload components, data ingestion module for processing multiple input datasets and remote scoring via FastAPI

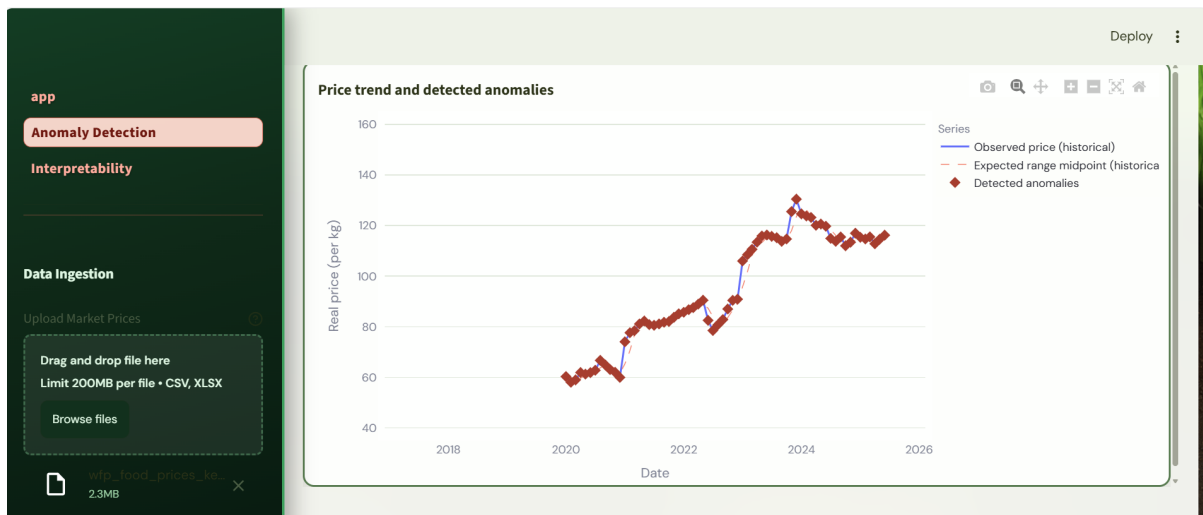


Figure 5.7: Price trend visualisation showing observed prices, expected baseline values, and detected anomalies over time

Figure 6.3 shows the distribution and proportion of anomaly severity levels, with high-risk cases dominating the detected anomalies.

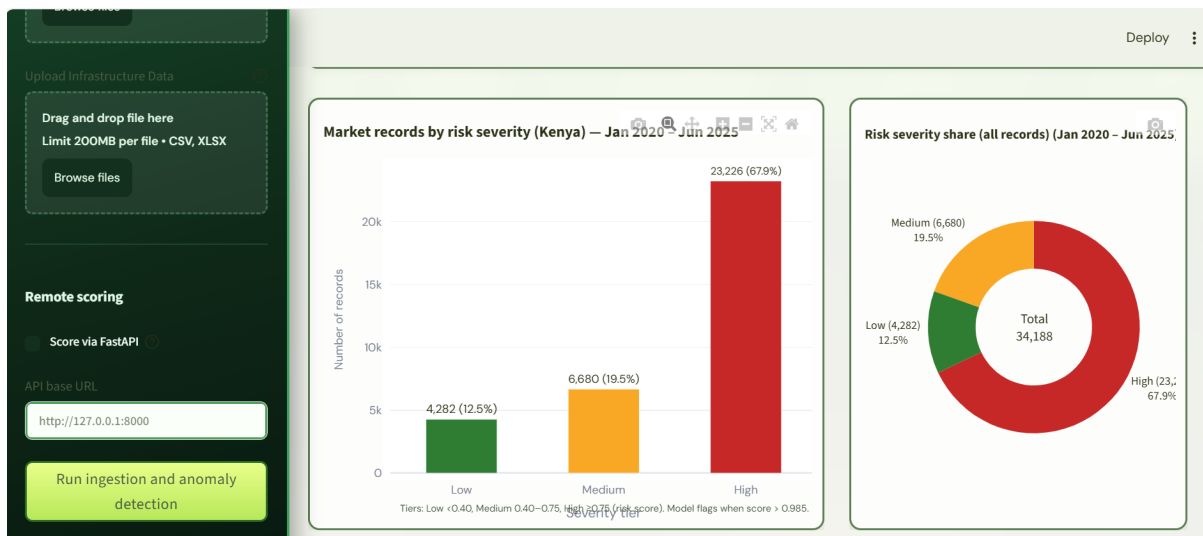


Figure 5.8: Distribution of detected anomalies by severity level, showing both absolute counts (bar chart) and proportional share (donut chart)

Figure 5.9 illustrates the monthly trend of anomaly severity levels. The figure shows trends in low, medium, and high-severity anomalies over time, with high-severity cases consistently dominating the anomaly profile.

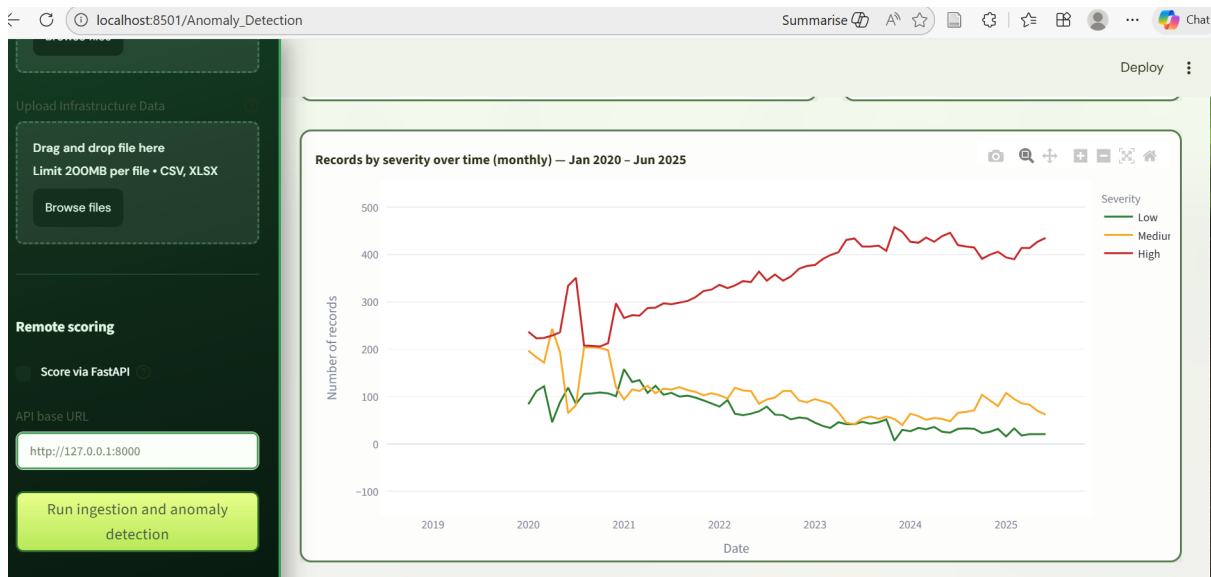


Figure 5.9: Temporal distribution of detected anomalies by severity level from January 2020 to June 2025.

5.2.3 Model Interpretability Interface

The model Interpretability interface provides insights into how the anomaly detection model makes predictions. This is achieved using SHAP-based explanations, which quantify the contributions on each feature to the model output.

The interface allows users to generate a global SHAP summary, which displays feature importance based on mean SHAP values. This enables identification of the most influential factors driving anomaly detection, such as price trends, lag variables or market conditions. Figure 5.10 presents the global SHAP feature importance, highlighting the key drivers of anomaly detection in the model. The plot ranks features based on their mean absolute SHAP values, indicating the relative contribution of each feature to anomaly prediction. Lagged price variables and rolling price trends are the most influential factors in detecting anomalies.

In addition, the interface supports local SHAP explanations for individual observations. Users can select a specific high-risk case and view how each feature contributes to the prediction. This provides a detailed understanding of why a particular price observation is classified as anomalous. Figure 6.9 illustrates the local SHAP explanation for a selected case, showing how individual features influence the anomaly prediction. Positive SHAP values indicate features pushing the prediction towards an anomaly, while negative values reduce the anomaly likelihood. The accompanying table provides the feature values and corresponding SHAP contributions for the selected instance.

These interpretability features enhance transparency and support informed decision-

making by allowing users to understand the underlying drivers of anomaly detection results.

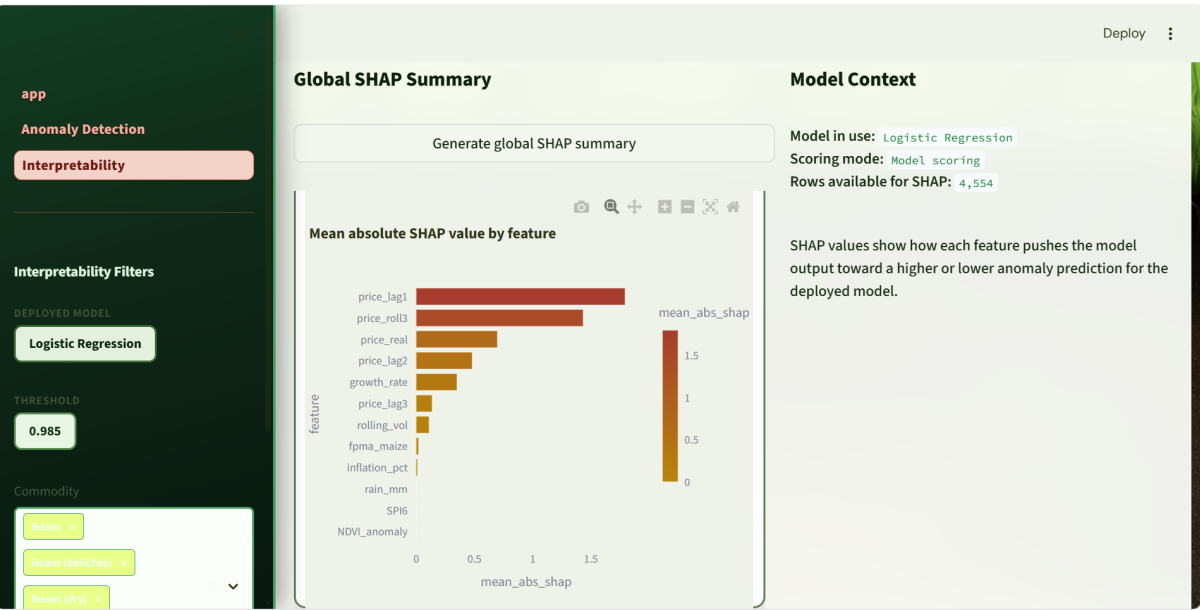


Figure 5.10: Global SHAP summary showing feature importance for the Logistic Regression model.

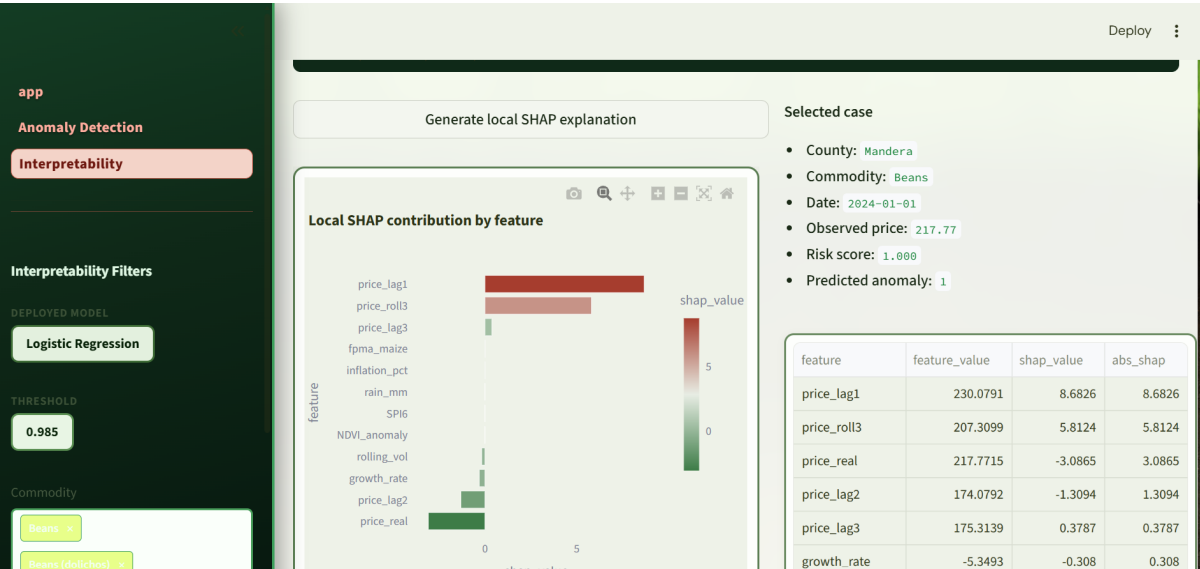


Figure 5.11: Local SHAP explanation for an individual anomaly case, showing the contribution of each feature to the model prediction.

5.3 System Testing

System testing was conducted to verify that the anomaly detection system operates correctly, reliably and as intended. The testing process focused on validating core functionalities, ensuring accurate execution and confirming that user interactions produce the expected outputs. Both functional and usability testing approaches were applied to assess system performance and user experience.

5.3.1 Functional Testing

Functional testing was performed to ensure that each system component executes its intended task without errors. This was done using separate data sourced from WFP Humanitarian database on food prices into the system for manual data ingestion and API for FastAPI ingestion. The key functionalities tested include data ingestion, model execution, dashboard interaction and output generation. Table 9 shows the results of functional testing.

Table 9: Functional Testing Results

Function	Test	Result
Dataset upload	Upload CSV file	Successful
Data preprocessing	Handle missing values	Successful
Model execution	Run anomaly detection	Successful
Dashboard filters	Change commodity/county	Charts updated correctly
Visualization	Display charts and maps	Successful
CSV export	Download alerts	Successful
API integration	FastAPI scoring request	Successful

The results show that all system functionalities operated as expected. Data uploads were successfully processed, the Logistic Regression model executed without errors and all visual outputs were updated dynamically based on user input. The integration with FastAPI also functioned correctly, enabling remote model scoring.

5.3.2 Usability Testing

Usability testing was conducted to evaluate how easily users can interact with the system and interpret its outputs. The system interface was designed with a simple and intuitive layout, allowing users to navigate between Dashboard, Anomaly Detection and Interpretability pages without difficulty.

The sidebar-based design provides clear access to filters such as commodity, county and date range, enabling users to refine analysis quickly. Interactive charts respond immediately to filter changes, improving responsiveness and user experience. Additionally, the use of visual elements such as KPI indicators, heatmaps and trend charts enhances the interpretability of results by presenting complex data in an accessible format.

Therefore, the system demonstrates good usability, with a clear structure, responsiveness, interactions and easily interpretable outputs that support effective anomaly monitoring and analysis.

5.4 Model Performance Evaluation

The evaluation of the Logistic Regression model used in the system for anomaly detection was based on standard classification metrics, including F1-score, Recall, and Area Under Curve (AUC), as well as visual performance analysis using the ROC curve and confusion matrix. The model achieved strong performance across all evaluation metrics, indicating its effectiveness in identifying anomalous food price patterns. Table 10 presents the performance of the Logistic Regression model, showing strong predictive capability with high AUC, balanced precision and recall, and a low false positive rate, indicating effective detection of anomalous food price behaviour.

Table 10: Logistic Regression Model Performance Metrics (ROC-Optimised Threshold)

Metric	Value
Precision	0.840
Recall	0.826
F1-score	0.833
True Positive Rate (TPR)	0.826
False Positive Rate (FPR)	0.160
AUC	0.916

The results show that the model achieves a high AUC of 0.916 (91.6%), indicating significant discrimination capability between normal and anomalous price observations. The F1-score of 0.833 (83.3%) reflects a balanced trade-off between precision and recall, while the recall of 0.826 (82.6%) shows that the model captures a substantial proportion of actual anomalies. The confusion matrices (TPR versus FPR) also show a reasonable balance between correctly detected anomalies and false alarms.

5.5 System Validation

The developed anomaly detection system effectively addresses the problem identified in Chapter 1, which is the need for timely and reliable detection of abnormal food commodity price behaviour. First, the system successfully detects abnormal price movements by applying a trained Logistic Regression model to historical and uploaded datasets. By identifying deviations from expected price patterns, the system enables early detection of potential market disruptions. Second, the system provides interpretable machine learning predictions through SHAP-based explanations. Equally, global and local interpretability features allow users to understand the factors driving anomaly detection, thereby increasing transparency and trust in the model outputs.

CHAPTER 6: DISCUSSION OF RESULTS

This chapter presents and discusses the results obtained from the food price anomaly detection system developed in this study. The focus is on interpreting the outputs generated by the deployed Logistic Regression model, particularly the anomalies identified across different commodities, time periods and counties. In addition, the chapter examines the performance and the insights provided by the model through the interpretability analysis.

The discussion is guided by the study objectives, namely: to develop a processing framework that captures temporal, spatial and inter-commodity dynamics; to train and evaluate supervised machine learning models for anomaly detection; and to assess the effectiveness of the developed system in supporting scalable and reliable food price monitoring. These objectives provide the basis for evaluating the detection results and the performance of the selected model.

The chapter further explores how anomalies vary over time and across regions, with attention to patterns in severity and distribution. These findings are interpreted in relation to underlying market conditions such as seasonality, climate variability and regional disparities. Thus, the results demonstrated how the developed system contributes to identifying abnormal price behaviour and supports a more data-driven approach to market surveillance.

6.1 Dataset Overview

The analysis is based on a consolidated dataset of food commodity prices and supporting variables covering 225 Kenyan markets across 27 counties. The dataset integrates monthly market price data sourced from WFP with complementary indicators, including rainfall, NDVI anomaly, and Standardised Precipitation Index (SPI), alongside engineered features such as price growth rates and rolling volatility. The data spans multiple commodities and counties, providing temporal depth and geographic coverage of anomaly detection.

The final dataset consists of 120,000 observations with price values expressed in real terms to account for inflation. The inclusion of climatic variables (rainfall, NDVI anomaly, SPI3 and SPI6) enabled the model to capture environmental influences on price behaviour, while derived features such as growth rate and rolling volatility are important in helping capture short-term price dynamics and fluctuations. This combination of variables was intended to support the model's capability in distinguishing between expected seasonal variations and abnormal price movements.

In addition, summary statistics show substantial dispersion across major variables. For instance, real prices range from 3.29 to 770.56 Kenya shillings per unit of food commodity, re-

flecting sharp increases and declines in prices over time. The relatively high rolling variability further confirms that price instability is present within the dataset. These characteristics highlight the complexity of food price behaviour and reinforce the need for a robust, data-driven anomaly detection approach. Table 11 shows the summary statistics of the key variables used in the anomaly detection model, highlighting the variability in price levels, climatic indicators and derived features.

Table 11: Summary Statistics of Key Variables

Variable	Mean	Std	Min	Max
price_real	59.81	53.33	3.29	770.56
NDVI_anomaly	0.00	0.06	-0.20	0.26
rain_mm	72.65	84.47	0.00	634.16
SPI3	0.00	1.00	-2.00	4.00
growth_rate	1.06	11.83	-81.10	705.14
rolling_vol	7.32	9.46	0.01	303.80

Figure 6.1 shows the distribution of real prices, indicating a right-skewed pattern with a concentration of observations at lower price levels and a long tail of higher values. This pattern is consistent with the wide range and variability reported in Table 11, confirming the presence of extreme price values within the dataset.

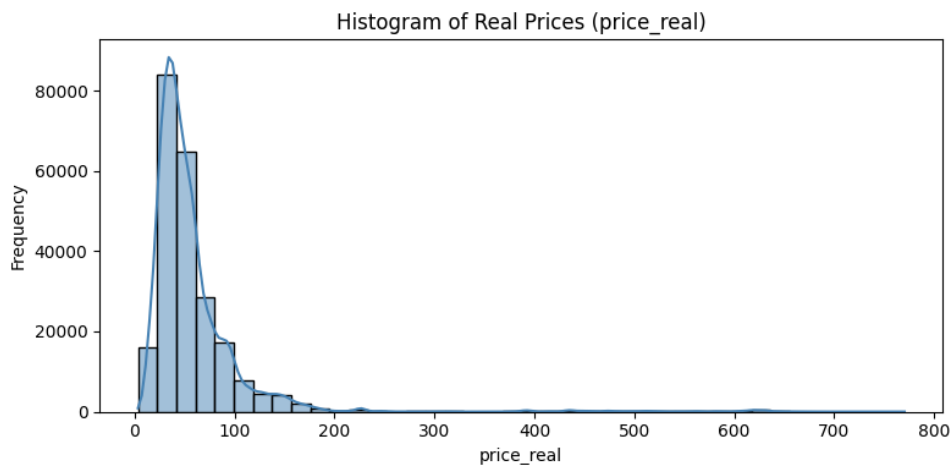


Figure 6.1: Distribution of real food prices (*price_real*) showing right-skewed behaviour with a concentration of observations at lower price levels and a long tail of high-price values.

6.2 Anomaly Detection Results

The results indicate that anomalies are present across the dataset and are not evenly distributed across observations. Instead, abnormal price behaviour is concentrated within specific commodities, time periods and geographic regions.

In terms of magnitude, a substantial number of anomalies were detected within the dataset, reflecting the presence of both gradual and sharp deviations from expected price behaviour. These anomalies were classified into three severity levels: low, medium and high. The classification provides a structured way of distinguishing between minor fluctuations and more significant disruptions in price patterns. Figure 3.6 illustrates the price trend alongside detected anomalies, highlighting periods where observed prices deviate significantly from expected patterns. The results show that anomalies are concentrated around sharp increases and declines in prices, suggesting that the model effectively captures sudden shocks and transitional deviations in market behaviour.

Price trend and detected anomalies



Figure 6.2: Price trend with detected anomalies, showing deviations of observed prices from expected patterns over time.

Figure 6.3 further reports on the distribution of anomalies across severity levels indicating that the majority of detected anomalies fall within the high-risk category. This implies that, the dominance of high-severity anomalies suggests that a large proportion of detected deviations represent significant departures from expected price behaviours rather than minor fluctuations.

Risk severity share (all records) (Jan 2020 – Jun 2022)

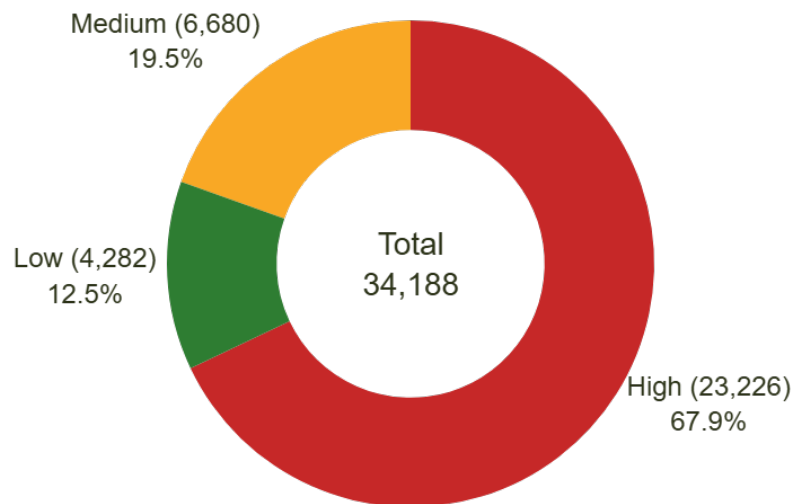


Figure 6.3: Distribution of detected anomalies by severity level, showing the proportion of low, medium, and high-risk observations.

6.3 Temporal Patterns of Price Anomalies

As indicated in Figure 6.2, the results indicate that anomalies are not randomly distributed across time but tend to occur in clusters, reflecting periods of significant deviations from expected price behaviour. Notably, the analysis of the temporal trend shows clear phases of increasing and decreasing anomaly activity. In particular, periods characterised by sharp increases in prices are associated with a higher concentration of anomalies. These spikes suggest the presence of market disruptions, where prices deviate rapidly from their expected range. Such behaviour is consistent with existing studies that link price volatility to shocks such as supply disruptions and climatic variability.

Figure 6.4 shows a clear upward trend in high severity anomalies over time, particularly from 2022 onwards. This indicates an increasing market instability, where a longer proportion of price movements deviate significantly from expected patterns. In contrast, low-severity

anomalies decline over time, suggesting that price behaviour is shifting from minor fluctuations towards more pronounced disruptions (Gilbert et al., 2017; FAO, IFAD, UNICEF, WFP and WHO, 2022).

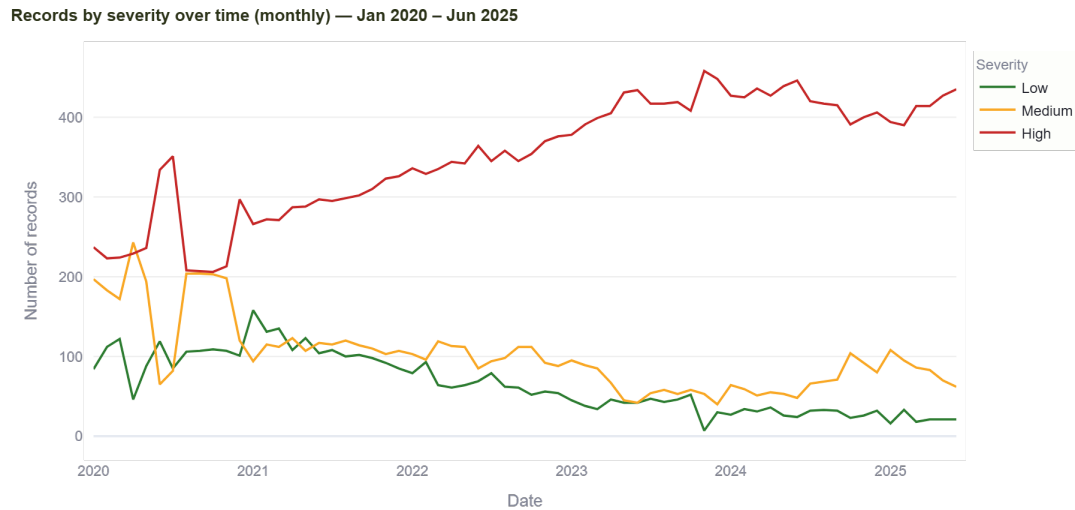


Figure 6.4: Monthly distribution of detected anomalies by severity level from January 2020 to June 2025.

Further, the clustering of anomalies around peak periods as illustrated in Figure 6.2 suggests that the model is sensitive to sudden shocks and sustained deviations in price behaviour. This aligns with findings in the literature that supervised machine learning models are effective in capturing nonlinear and time-dependent patterns in commodity prices (Erfani et al., 2016; Chen and Guestrin, 2016). Similarly, the temporal distribution of anomalies also reflect underlying seasonal and external influences. The variations in anomaly intensity overtime is linked to factors like climatic conditions, harvest cycles and market supply constraints, which are know to influence food price dynamics (Zhang et al., 2025; Waiswa, 2023). The ability of the model to detect anomalies during these periods highlights its importance in identifying early signals of market stress.

Consequently, the temporal analysis demonstrates that anomaly detection provides meaningful insights into when market instability occurs. The system supports timely monitoring and early intervention by identifying periods of heightened anomaly activities. This addresses the study objective of developing a data-driven framework capable of capturing temporal dynamics in food price behaviour for improved market surveillance.

6.4 Spatial Distribution of Anomalies

The results indicate that anomalies are not evenly distributed spatially but are concentrated in specific regions, reflecting differences in local market conditions and structural factors. The

spatial analysis reveals that certain counties consistently record higher levels of anomaly activity, particularly in the high-severity category. This suggests that these regions experience more pronounced deviations in price behaviour, which may be linked to factors such as limited market access, higher transport costs or exposure to climatic shocks. In contrast, other counties exhibit relatively lower levels of anomaly activity, indicating more stable price patterns.

Figure 6.5 illustrates the geographic distribution of anomalies, highlighting clusters of high-risk areas. The observed clustering suggests that price instability is not isolated but may affect neighbouring regions with similar economic or environmental conditions. This pattern is consistent with existing literature, which shows that spatial factors such as infrastructure and trade connectivity play a significant role in shaping food price dynamics (Waiswa, 2023; Kihiu and Gachanja, 2022).

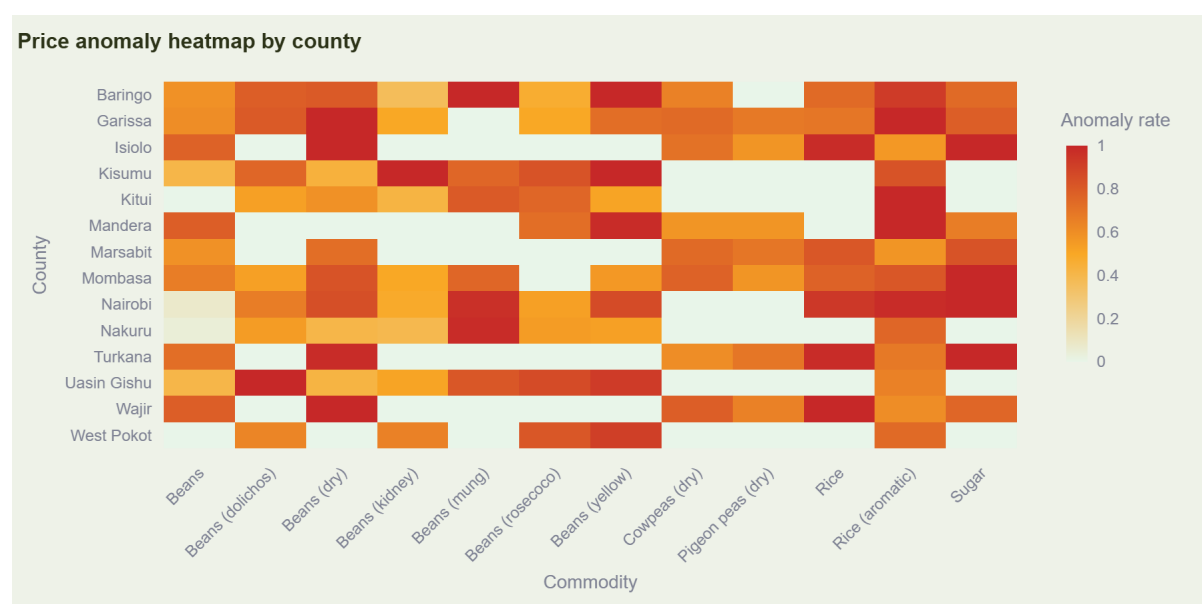


Figure 6.5: Spatial distribution of anomaly rates across counties and commodities, highlighting variations in market instability across regions.

The presence of regional disparities in anomaly distribution underscores the importance of incorporating spatial features in anomaly detection models. By capturing these differences, the system provides more targeted insights into where market disruptions are likely to occur. This supports more focused monitoring and intervention strategies, particularly in high-risk regions. As a result, the spatial analysis revealed that anomaly detection can effectively identify geographic patterns of market instability, supporting the study objective of incorporating spatial variation into a data-driven framework for food price monitoring.

6.5 Model Performance Analysis

Several supervised machine learning models were evaluated for anomaly detection, including Logistic Regression, Linear SVM, XGBoost, LightGBM and Decision Tree. Their performance was assessed on the ability to discriminate normal from anomalous price observations while minimising false detection, leading to the selection of the most suitable model for deployment within the system.

The initial evaluation using stratified cross-validation, illustrated in Table 12, showed that Linear SVM and Logistic Regression achieved the most consistent performance across precision, recall and AUC. In contrast, the XGBoost and LightGBM achieve quite high recall but at the expense of lower precision, indicating a tendency to over-identify anomalies. This highlights the importance of balancing detection sensitivity with false alarm control in anomaly detection tasks (Aggarwal, 2017; Erfani et al., 2016).

Table 12: Model Performance under Stratified 5-Fold Cross-Validation (mean $\pm$ std)

Model	Precision	Recall	F1-score	ROC-AUC
Linear SVM	0.718 $\pm$ 0.044	0.948 $\pm$ 0.021	0.816 $\pm$ 0.025	0.886 $\pm$ 0.023
XGBoost	0.606 $\pm$ 0.038	0.993 $\pm$ 0.005	0.752 $\pm$ 0.028	0.888 $\pm$ 0.028
Logistic Regression	0.723 $\pm$ 0.035	0.929 $\pm$ 0.044	0.811 $\pm$ 0.015	0.885 $\pm$ 0.021
LightGBM	0.610 $\pm$ 0.040	0.992 $\pm$ 0.004	0.755 $\pm$ 0.030	0.887 $\pm$ 0.029
Decision Tree	0.743 $\pm$ 0.038	0.827 $\pm$ 0.124	0.776 $\pm$ 0.049	0.830 $\pm$ 0.035

Further improvement on the operational performance of the models was implemented by optimising classification thresholds using ROC analysis based on Youden’s J statistic. The resulting optimal thresholds were high across the models, with Logistic Regression (0.988) and boosting models (0.998) requiring stricter decision boundaries to control false positives. This reflects the imbalanced nature of the dataset, where anomalies are rare and require careful threshold selection to avoid excessive false alarms.

Table 13: Optimised Model Performance using ROC-based Thresholds

Model	Precision	Recall	F1-score	FPR	AUC
Logistic Regression	0.849	0.819	0.834	0.147	0.919
Linear SVM	0.840	0.830	0.835	0.161	0.918
XGBoost	0.842	0.763	0.801	0.145	0.897
LightGBM	0.854	0.722	0.782	0.126	0.889
Decision Tree	0.724	0.841	0.778	0.325	0.793

The optimised results show that Logistic Regression achieved the highest AUC (91.9%) and maintains a balance between precision (84.9%) and recall (81.9%), resulting in a stable F1-score (83.4%). Linear SVM showed comparable performance, while XGBoost and LightGBM demonstrated reduced recall after threshold optimisation, confirming that their earlier high recall was partly driven by less restricted thresholds. The Decision Tree model records the weakest overall performance, particularly because of a higher false positive rate.

Further, Figure 6.6 confirms the strong discriminative ability of Logistic Regression and Linear SVM, while the confusion matrices in Figure 6.7 further indicate that Logistic Regression achieves a balanced classification outcome, correctly identifying anomalies while maintaining controlled false positives. In contrast, boosting models exhibited a tendency to over-predict anomalies, consistent with their high recall scores.

While most models demonstrated strong detection capability, Logistic Regression provided the most balanced and stable performance under both cross-validation and threshold-optimised conditions, thereby emerging as the most suitable model for deployment within the anomaly detection system (Santoro et al., 2024).

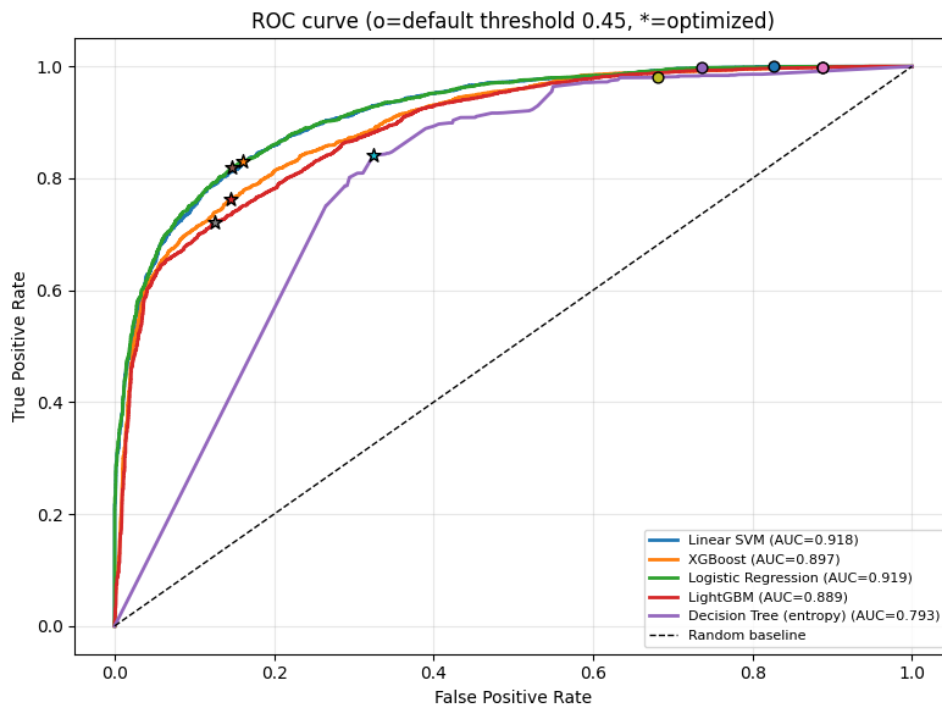


Figure 6.6: ROC curves comparing the performance of different models in distinguishing normal and anomalous observations.

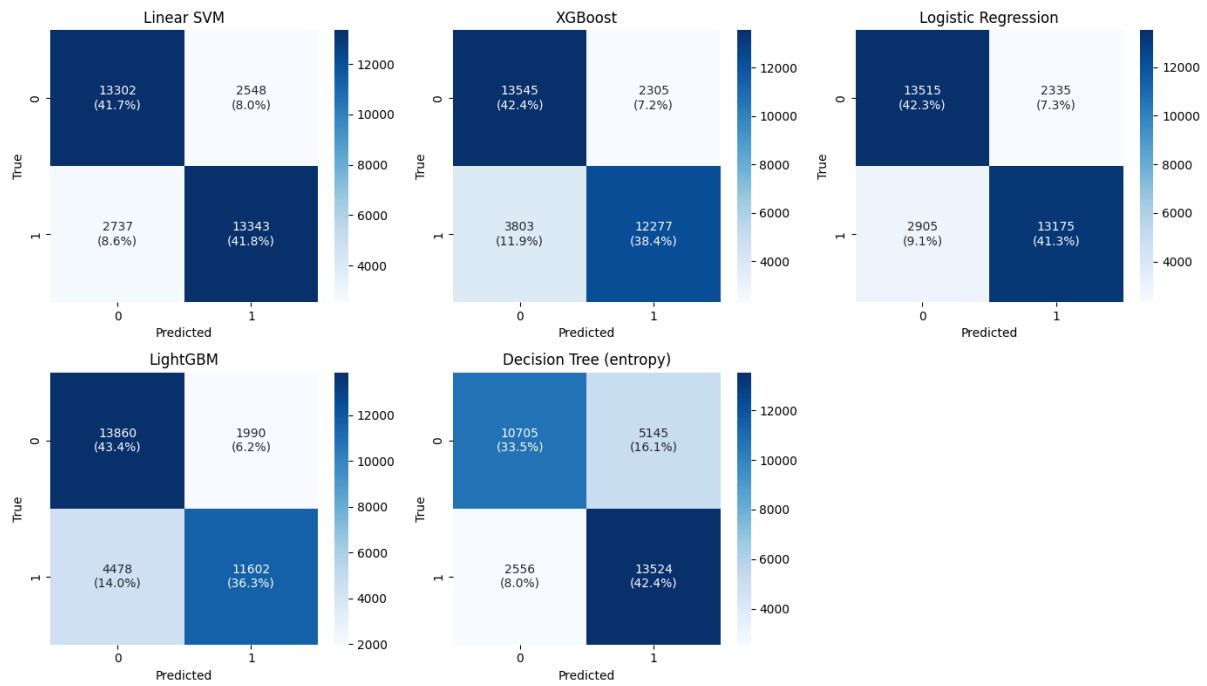


Figure 6.7: Confusion matrices for evaluated models, showing classification performance across normal and anomalous classes.

6.6 Model Interpretability (SHAP Analysis)

Understanding how the selected model makes predictions is important for ensuring transparency and practical usability. In this study, SHAP (SHapley Additive exPlanations) values were used to explain the contribution of each feature to anomaly detection. This allows the model's decisions to be interpreted in terms of real-world drivers of food price instability rather than treated as black box.

Figure 6.8 shows the global importance of features across all price observations. The results indicate that lagged price variables (`price_lag1`, `price_roll3` and `price_lag2`) are the most influential predictors of anomalies. This suggests that recent price movements and short-term trends are the strongest signals of abnormal behaviour. Growth rate and rolling volatility also contribute meaningfully, reinforcing that sudden changes and instability also contribute meaningfully, reinforcing that sudden changes and instability in prices are major indicators of anomaly. In contrast, climatic variables such as rainfall, NDVI anomaly and SPI show relatively lower contributions, indicating that their effects are more indirect or captured through price dynamics.

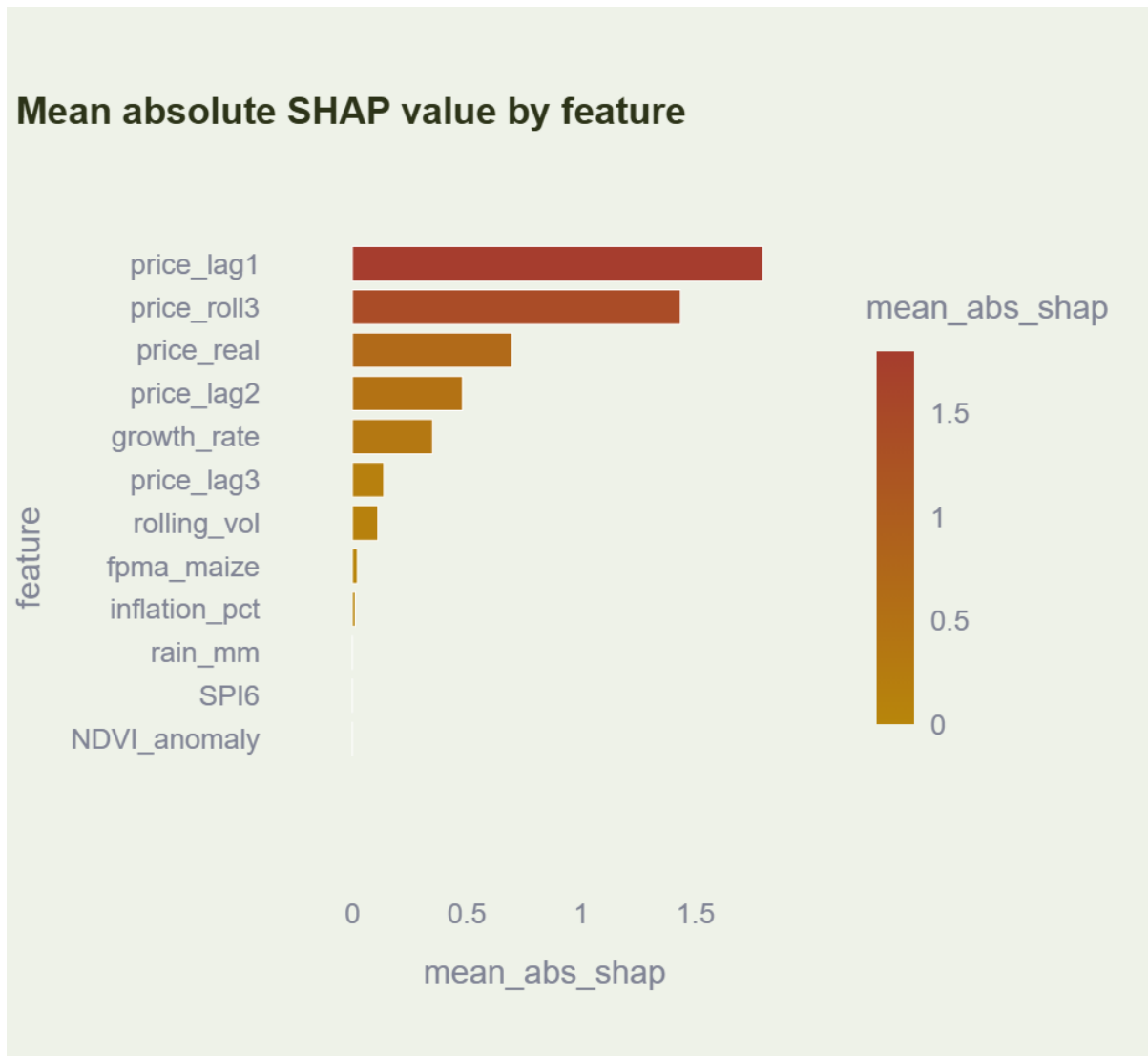


Figure 6.8: Global feature importance based on mean absolute SHAP values

Figure 6.9 presents local explanation for an individual anomaly case. The results show that high values of recent price indicators (textttprice_lag1 and price_roll3) strongly push the prediction toward anomaly, while current price levels (price_real) and certian lagged values contribute negatively, moderating the prediction. This demonstrates that anomaly detection is driven by deviations from recent trends rather than absolute price levels alone.

Thus, the interpretability analysis shows that the model relies primarily on short-term price dynamics, supported by volatility and growth indicators, to identify anomalies. This aligns with the existing literature, which highlights that food price anomalies are often triggered by rapid market shifts rather than gradual environmental changes (Gilbert et al., 2017; Zhang et al., 2025). Consequently, providing both global and local explanations implies that the model not only achieves predictive performance but also offers interpretable insights that

can support decision-making in food price monitoring systems.

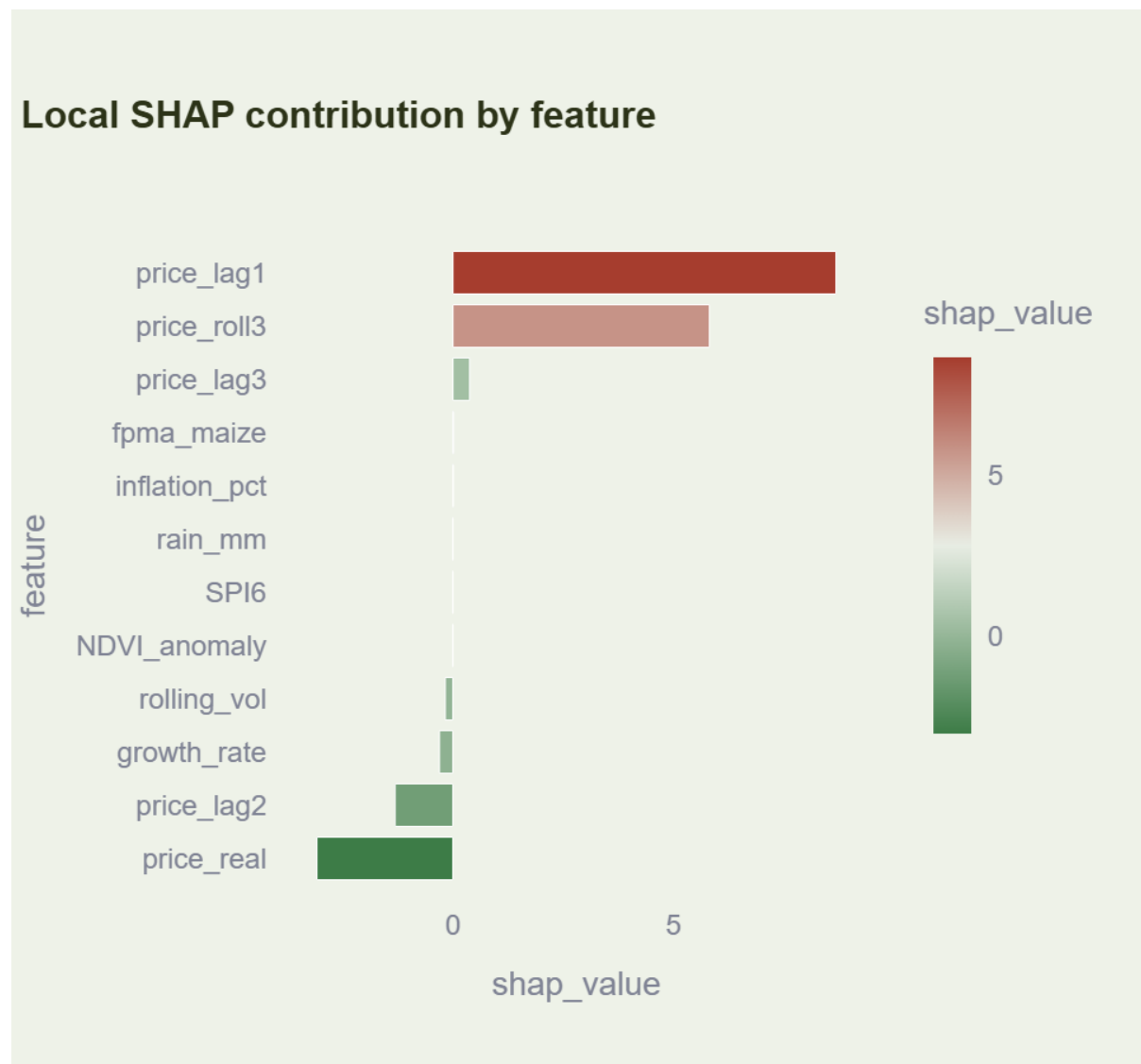


Figure 6.9: Local SHAP explanation for a selected anomaly observation

6.7 Implications for Food Price Monitoring

The findings of this study demonstrated that supervised machine learning techniques can significantly improve the detection of anomalies in food commodity prices. This is exhibited by the capability of the system to provide a more responsive and data-driven alternative by capturing short-term price dynamics, volatility and temporal patterns, compared to traditional monitoring approaches which often rely on fixed thresholds and delayed reporting.

One major implication is the ability to support early warning systems. The temporal analysis shows that anomalies tend to cluster during specific periods indicating episodes of market instability. By identifying these periods in real-time, the system can provide timely signals to policymakers and market stakeholders, enabling earlier interventions to stabilise prices

and mitigate food security risks.

Moreover, the spatial analysis further highlights that anomaly patterns vary across countries, reflecting differences in market access, supply conditions and local shocks. This suggests that a uniform monitoring approach may not be sufficient and that targeted, region-specific strategies are necessary. The systems's ability to identify geographic hotspots of instability can support more focused policy responses.

From operational perspective, the model selection results show that a relatively simple and interpretable model, such as Logistic Regression, can achieve strong and reliable performance. This has practical advantages, including ease of implementation, lower computational requirements and greater transparency compared to more complex models. As a result, the system is well suited for deployment in resource-constrained environments where scalability and interpretability.

Therefore, the study demonstrates that integrating machine learning into food price anomaly systems enhances accuracy and timeliness of anomaly detection. This can support proactive and evidence-based decision-making, contributing to improved market stability and resilience.

CHAPTER 7: CONCLUSIONS, RECOMMENDATIONS AND FUTURE WORKS

This chapter presents the conclusions of the study on detecting anomalies in food commodity prices using supervised machine learning techniques. It summarises the main findings in relation to the research objectives and interprets their implications for food price monitoring. The chapter also outlines practical recommendations based on the results and highlights areas for further research to improve and extend the proposed framework.

7.1 Summary of Key Findings

The findings of this study are presented in relation to the three research objectives, with particular emphasis on how some of the identified gaps in existing approaches were addressed.

For the first objective, the study developed a data engineering framework that integrates temporal, spatial and climatic features, addressing the gap in limited feature integration identified in the literature. The results show that lagged price variables, rolling averages and growth rates are the most influential in detecting anomalies, indicating that short-term price dynamics are the primary drivers of abnormal market behaviour. While climatic indicators such as rainfall, NDVI, and SPI were included to capture environmental effects, their relatively low contributions suggest that these factors influence prices indirectly through market dynamics.

For the second objective, multiple supervised machine learning models were evaluated, including Logistic Regression, Linear SVM, XGBoost, LightGBM, and Decision Tree, to address the reliance on traditional threshold-based methods. The findings showed that Logistic Regression achieved the most balanced and consistent performance, particularly after threshold optimisation. It maintained a strong trade-off between precision and recall, achieved a high AUC and demonstrated stability across validation folds. In contrast, boosting models exhibited very high recall but lower precision, underscoring the importance of balancing detection sensitivity with false-alarm control in practical applications.

For the third objective, the study demonstrates that the proposed framework is scalable and adaptable, addressing the literature on the transferability of models across different market contexts. The use of structured features and a relatively simple, interpretable model makes deployment easier and reduces computational complexity. The framework can therefore be applied to other regions with similar data structures, requiring only retraining rather than redesign.

The findings demonstrate that the study objectives have been achieved, particularly in addressing gaps related to feature integration, model reliability and scalability. The proposed approach offers an interpretable, practical framework for detecting food price anomalies,

demonstrating its potential to support real-world food price monitoring.

7.2 Conclusions

This study aimed to develop a supervised machine learning framework for detecting anomalies in food commodity prices and address key limitations in existing monitoring approaches. The findings show that food price anomalies are primarily driven by short-term price dynamics, particularly changes captured by lagged prices, growth rates, and volatility measures. These findings demonstrated the importance of temporal features in understanding abnormal market behaviour.

The evaluation of multiple models demonstrated that supervised machine learning can capture complex and nonlinear patterns in food price data. Logistic Regression, in particular, showed stable, balanced performance, making it suitable for practical deployment. Its consistency across validation and threshold optimisation confirms that reliable anomaly detection does not necessarily require highly complex models, especially when supported by well-engineered features.

The inclusion of spatial and climatic variables further strengthens the framework by providing additional context to price movements. While these factors were not dominant drivers in the model, they contribute to a more comprehensive understanding of the conditions under which anomalies occur. This supports a more holistic approach to food price monitoring beyond purely price-based analysis.

More so, the study demonstrates that anomaly detection can provide meaningful insights into when and where market instability occurs. Temporal patterns reveal periods of heightened anomaly activity, while spatial analysis highlights regional disparities across counties. These insights are critical for improving monitoring systems and enabling more targeted responses to market disruptions.

To recap, these findings show that the developed framework meets the study objectives and provides a practical basis for improving food price monitoring with the potential to support efforts aimed at enhancing market stability in Kenya and beyond in regions with similar market conditions.

7.3 Recommendations

Based on the findings of this study, several recommendations are proposed to enhance food price monitoring and support effective responses to market stability.

At the policy level, national institutions such as the Kenya National Bureau of Statis-

tics (KNBS) and the Ministry of Agriculture can integrate data-driven anomaly detection approaches into existing monitoring systems. At the regional and global level, organisations such as the World Food Programme (WFP) and the Food and Agriculture Organisation (FAO) can incorporate machine learning-based tools into their early warning frameworks, such as ALPS and FPMA, to complement traditional methods and improve the timeliness and accuracy of food price anomaly detection.

From a technical perspective, improving data quality and integration remains critical. Stakeholders require investment in harmonising datasets across sources, reducing missing observations and incorporating additional variables like market connectivity, trade flows and transport costs. Strengthening real-time data pipelines and data-sharing mechanisms between national and international agencies would further enhance the responsiveness and scalability of the system.

Operationally, the developed system can be deployed as a decision-support tool for policymakers, analysts and humanitarian agencies. Institutions such as the WFP, FAO and regional bodies can use the system to monitor market conditions, identify emerging risks and support targeted interventions. Equally, capacity building is an essential component, particularly in training users to interpret anomaly signals and integrate model outputs into the decision-making process.

At a broader level, the adoption of data-driven monitoring frameworks contributes to global development priorities, particularly the Sustainable Development Goals (SDGs). In particular, the approach supports SDG 2 (Zero Hunger) by enabling early detection of food price instability, and SDG 13 (Climate Action) by incorporating climate-related indicators into monitoring systems. Strengthening collaboration between local, regional and global stakeholders will be key to ensuring that such frameworks are effectively implemented and sustained.

Therefore, these recommendations emphasise the need for coordinated efforts across institutions to improve food price monitoring systems, leveraging machine learning to enhance early warning capabilities and support market stability.

7.4 Limitations of the Study

This study is subject to several limitations that should be considered when interpreting the results.

First, the analysis relied on secondary data sources, which contain missing values, inconsistencies and varying levels of spatial and temporal coverage. Although data preprocessing

techniques were applied to address these issues, such limitations may still affect the accuracy of the model.

Second, the availability of labelled anomaly data is limited, requiring the use of proxy or derived anomaly labels. This may introduce uncertainty in the training process, as the labels may not fully capture all real-world anomaly conditions. Consequently, the performance of the model may vary when applied to unseen or more complex market scenarios.

Third, while climatic and economic variables were incorporated, their representation may not fully capture the complexity of market dynamics. The economic indicators, such as inflation and global price trends, are often aggregated and may not reflect local market conditions, while other relevant factors, such as transport cost, trade restrictions and market integration, were not fully included. This has the potential to limit the ability of the model to account for all drivers of price variability.

Finally, the performance of the model may be affected during extreme or rare market shocks, where patterns differ significantly from historical trends. As with many supervised learning approaches, the model depends on historical data and may be less effective in capturing unprecedented events.

Irrespective of these limitations, the study provides a robust framework for anomaly detection and offers a strong foundation for further improvements in food price monitoring systems.

7.5 Future Work

Building on the findings and limitations of this study, several areas for future research and development are identified to further enhance the proposed framework.

First, future work can focus on incorporating real-time data streams to improve the timeliness of anomaly detection. Integrating live data from market information systems, remote sensing platforms and economic indicators would enable continuous monitoring and support more responsive early warning systems.

Second, alternative modelling approaches can be explored to complement the supervised learning framework. Semi-supervised and unsupervised anomaly detection techniques may help address the challenge of limited labelled data and improve the detection of rare or previously unseen anomalies.

Third, the inclusion of additional variables should strengthen the explanatory power of the model. Future studies can incorporate more detailed economic and market structure

indicators, such as transport costs, trade flows, market connectivity and policy interventions, to better capture the drivers of price variability.

Fourth, spatial modelling can be further enhanced by incorporating advanced techniques such as spatial econometrics or graph-based models. These approaches would allow for a more explicit representation of relationships between markets and improve the detection of geographically driven anomalies.

Finally, the framework can be extended and validated across different countries and regions to assess its generalisability. Applying the model to other food markets beyond Kenya would provide insights into its adaptability and support the development of scalable, cross-country monitoring systems.

These directions provide a pathway for improving both the analytical capability and applicability of machine learning-based food price anomaly detection systems.

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APPENDICES

Appendix A: Data Description and Sources

This appendix provides detailed information on the datasets used in the study.

Food Price Data

The primary dataset was obtained from the World Food Programme (WFP) price database via OCHA, covering monthly food commodity prices across Kenyan markets from 2006 to 2025. The dataset includes 137,358 observations across 47 counties and major commodities such as maize, beans, and rice.

Climate Data

Climate variables were derived from Google Earth Engine, including NDVI (MODIS) and rainfall (CHIRPS), aggregated at the county level. These variables capture vegetation health and precipitation patterns relevant to agricultural output.

Economic Data

Macroeconomic indicators were sourced from the Kenya National Bureau of Statistics (KNBS), including Consumer Price Index (CPI) and inflation rates, used to adjust nominal prices into real values.

Spatial Data

County boundary shapefiles were obtained from FAO GAUL and used for spatial aggregation and analysis.

Appendix B: Data Preprocessing

Missing Value Handling

Missing values were addressed using a combination of forward and backwards filling for temporal continuity and KNN imputation for multivariate features.

Inflation Adjustment

Nominal prices were converted to real prices using CPI-based deflation, with 2015 as the base year.

Anomaly Labelling

Anomalies were defined using a rolling percentile approach (70th percentile over a 6-month window), with a fallback global threshold where necessary.

Appendix C: Feature Engineering

Temporal Features

Temporal dynamics were captured using growth rates, rolling volatility, lagged prices, and rolling averages.

Climate Features

Climate indicators included NDVI anomalies, rainfall, and SPI (Standardised Precipitation Index), along with their lagged values.

Inter-Commodity and Economic Features

Global commodity prices and inflation rates were incorporated to capture external market influences.

Feature Summary

Table 14: Summary of Engineered Features

Feature	Category	Description
price_real	Price	CPI-adjusted price
growth_rate	Temporal	Monthly price change
rolling_vol	Temporal	Price volatility
NDVI_anomaly	Climate	Vegetation deviation
SPI3	Climate	Drought index
inflation_pct	Economic	Inflation rate

Appendix D: Model Development and Tuning

Train-Test Split

A temporal split was used, with training data from 2006–2022 and testing data from 2023–2025.

Cross-Validation

An expanding window approach was used to preserve temporal ordering during validation.

Hyperparameter Tuning

Models were tuned using GridSearchCV and RandomizedSearchCV with recall-focused scoring due to class imbalance.

Threshold Optimization

Optimal classification thresholds were determined using ROC curves and Youden’s J statistic.

Appendix E: Model Implementation

Pipeline Execution

The system follows a structured pipeline: data loading → preprocessing → feature engineering → model training → evaluation.

Model Persistence

Trained models and preprocessing components were stored using joblib for deployment.

Appendix G: Model Interpretability

SHAP (SHapley Additive Explanations) was used to interpret model predictions, providing both global and local feature importance insights.

Appendix H: Code Snippets

This section provides selected code excerpts illustrating key implementation steps.

Feature Engineering Example

```
df['growth_rate'] = df['price'].pct_change()  
df['rolling_vol'] = df['price'].rolling(3).std()
```

Model Loading

```
model = joblib.load('model.pkl')
```

Appendix I: System and Environment Setup

Software Requirements

Python 3.11+ with key libraries including pandas, scikit-learn, XGBoost, LightGBM, and SHAP.

Performance Considerations

Memory optimisation and parallel processing were applied to improve efficiency during model training.